Kullback–Leibler Divergence-Aware Dimensionality Reduction on HPD Manifold: Enhancing Feature Discrimination for Multichannel SAR Moving Target Detection

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***Abstract*—Clutter suppression remains a fundamental challenge in multichannel synthetic aperture radar (SAR) moving target detection. Among various approaches, space–time adaptive processing (STAP) has been widely used due to its excellent clutter suppression capabilities. However, STAP intrinsically assumes complex Gaussian clutter, which often significantly deviates from actual heterogeneous clutter environments encountered in practical scenarios. Such deviations can markedly impair the clutter suppression performance of STAP. To overcome this issue, this article proposes a Kullback–Leibler divergence (KLD)-based Hermitian positive definite (HPD) manifold dimensionality reduction (DR) technique, aimed at enhancing the feature discrimination of moving target on HPD manifold. Initially, the airborne multichannel SAR image domain data are mapped into the HPD manifold to form a feature manifold. Next, an optimal DR mapping is constructed by formulating an optimization problem on the HPD manifold that maximizes the KLD, and an analytical solution to this problem is derived. Based on this solution, an optimal reduction matrix is designed, projecting the data into a low-dimensional HPD manifold. Finally, on the reduced HPD manifold, a matrix information geometry (MIG) detector based on KLD is used to achieve enhanced detection of moving targets. The proposed multichannel SAR moving target detection framework incorporating KLD-based DR does not rely on specific parametric statistical model of the clutter background, and only uses geometric properties derived from second-order statistic. This effectively addresses the performance degradation caused by statistical distribution model mismatch in complex environments. Moreover, deriving an analytical solution to the optimization problem avoids the high computational costs associated with Riemannian conjugate gradient algorithms, thus reducing complexity substantially and facilitating practical implementation in SAR systems. Experimental results, including simulations and measurements with real clutter backgrounds, demonstrate the superior effectiveness of the proposed approach.**

***Index Terms*—Clutter suppression, dimensionality reduction (DR), Hermitian positive definite (HPD) manifold, Kullback–Leibler divergence (KLD), synthetic aperture radar (SAR) moving target detection.**

Received 22 May 2025; revised 24 July 2025; accepted 15 August 2025. Date of publication 18 August 2025; date of current version 28 August 2025. This work was supported in part by the Science Fund for Distinguished Young Scholars of Hunan Province under Grant 2024JJ2066, in part by the Science and Technology Innovation Program of Hunan Province under Grant 2022RC1092, and in part by China Postdoctoral Science Foundation under Grant 2021TQ0084. *(Corresponding author: Yaowen Fu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3599910

I. Introduction

SYNTHETIC aperture radar (SAR) ground moving target indication (GMTI) techniques can extract high-precision information related to moving targets from SAR data, playing a crucial role in many civilian and military applications [1], [2], [3], [4], [5], [6]. However, moving targets undergo azimuthal displacement and defocusing after SAR imaging processing [7], [8]. Moreover, the movement of the platform causes clutter spectrum expansion, making it difficult to detect slow-moving targets submerged in strong clutter [9]. Therefore, clutter suppression is a key challenge in SAR-GMTI processing. Multichannel SAR systems leverage multiple antennas along the azimuth direction to provide additional spatial information for the signal, enabling joint processing for clutter suppression, which offers superior performance compared with single-channel configurations and enhances the detection capabilities for slow and weak-moving targets. Currently, space–time adaptive processing (STAP) is widely used in multichannel SAR systems. In complex Gaussian clutter environments, based on the criterion of maximizing the signal-to-clutter-noise ratio (SCNR), it can achieve the theoretically optimal clutter suppression performance [10], [11], [12]. SAR-STAP processing requires estimating the clutter covariance matrix (CCM) of the cell under test (CUT) using several independent and identically distributed (IID) training cells [13]. A commonly used technique is the sampling matrix inversion (SMI). The SMI estimator is inherently the maximum likelihood estimate (MLE) under the complex Gaussian (homogeneous) clutter assumption. Moreover, STAP can be modeled as a mean-shifted Gaussian–Gaussian problem under the complex Gaussian hypothesis, and the generalized likelihood ratio test (GLRT) derived from the complex Gaussian model yields a test statistic equivalent to the optimal STAP weight vector [9], [14]. However, the real-world detection scenarios are often heterogeneous, involving terrain undulations, forests, roads, bridges, buildings, and other artificial structures. These features cause significant power variations across range cells and violate the complex

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Gaussian distribution assumption for clutter, leading to a significant degradation of STAP performance and a marked decline in clutter suppression efficacy [15]. For heterogeneous environments, a more general clutter model is the compound Gaussian model [16], [17], which represents clutter as the product of a fast-changing texture component and a slowly varying speckle component. Although statistics-based optimal detectors can be derived for different texture distributions, the mismatch between real clutter environments and the assumed statistical model results in considerable performance degradation of these model-dependent detection methods, making them less suitable for practical detection tasks.

In recent years, radar signal processing techniques driven by information geometry (IG) have attracted extensive attention and played important roles in fields such as radar target detection [18], [19], [20], [21], [22] and recognition [23]. Currently, matrix IG (MIG) detectors are widely used in radar target detection [24], [25]. Radar adaptive target detection techniques based on MIG can fully use the nonlinear geometric structure of the signal itself. Different from methods based on statistical distribution models, MIG detectors do not rely on specific parametric statistical model of clutter, and only use geometric properties derived from second-order statistic, providing strong support and new technical approaches for the nonlinear processing of radar signals. This is expected to improve the detection performance in heterogeneous clutter environments. The MIG detectors construct covariance matrices for each data, map them into the Hermitian positive definite (HPD) manifold, and then calculate the geometric distance between the covariance matrices of CUT and the clutter center to make detection decisions. Initially, the MIG detector used the affine invariant Riemannian metric (AIRM) as the distance metric function. However, the calculation of the AIRM and its geometric mean involves the matrix logarithm and the iterative solution of the fixed-point method, resulting in a large computational cost. To improve computational efficiency, scholars have begun to search for more efficient distance metric functions. The log-Euclidean metric (LEM) [21] and information divergences have been applied to MIG detectors [26], greatly expanding the types of available distance functions. Among these geometric metrics, the Kullback–Leibler divergence (KLD) has excellent detection performance and local equivalence to the AIRM, making it the most typical and widely used information divergence [27], [28].

Currently, research on MIG detectors mainly focuses on improving the distance metric function, while relatively few studies focus on optimizing the original feature manifold structure of the data. According to the manifold distribution law, high-dimensional data of the same category in nature often concentrate on a certain low-dimensional manifold, making data of the same category to be concentratedly distributed [29]. Harandi et al. [30] point out that dimensionality reduction (DR) processing through manifold transformation on the HPD Riemannian manifold may increase the feature differences in data from different categories, thereby expanding their geometric distances and making them more distinguishable. Typical DR processing techniques, such as principal component analysis (PCA) [31] and linear discriminant analysis (LDA) [32], are widely used in the flat Euclidean space. However, they ignore the nonlinear structure of the data itself and are thus not suitable for DR processing on the HPD Riemannian manifold. Manifold learning methods that consider the structure of the Riemannian manifold itself, such as isometrical mapping (Isomap) [33], locally linear embedding (LLE) [34], and Laplacian eigenmap (LE) [35], have been proposed successively. However, such manifold learning methods are mostly used in transductive learning tasks and it is difficult to directly obtain the explicit mapping of the low-dimensional manifold. Yang et al. [36] fully consider the geometric structure of the HPD manifold and combine feature dimension reduction with radar adaptive detection, effectively improving the detection performance. However, this method requires calculating the Riemannian gradient in each iteration, resulting in a large computational cost. For SAR systems with a large amount of data, it is difficult to directly apply the method to multichannel SAR moving target detection.

Based on the foregoing discussions, designing effective and computationally tractable DR methods on the HPD Riemannian manifold remains a challenge. To address the challenges of moving target detection in multichannel SAR systems under heterogeneous clutter backgrounds and overcome the high computational complexity of existing manifold DR methods, this article proposes an HPD Riemannian manifold DR mapping method based on the KLD, aiming to enhance the feature discrimination of moving target on HPD manifold. First, the data in the airborne multichannel SAR image domain are represented as points on the HPD manifold through feature mapping, constructing the feature manifold. Then, the geometric mean of the training samples is used to estimate the CCM, and the CUT is used to estimate the covariance matrix under the $\mathcal{H}_1$ hypothesis (target present). Subsequently, we formulate a DR mapping optimization problem based on maximizing the KLD and derive the analytical solution of this optimization problem, thus avoiding the huge computational cost of using the Riemannian conjugate gradient method. Based on this analytical solution, we construct the optimal DR matrix and project the data into the low-dimensional HPD manifold. Finally, on the low-dimensional HPD manifold, a KLD-based MIG (KLD-MIG) detector is used to detect moving targets. The multichannel SAR moving target enhanced detection framework based on KLD DR mapping proposed in this article significantly reduced the computational complexity, making it suitable for processing in SAR systems. The experimental results show that compared with the traditional STAP method and the KLD-MIG detector without DR mapping, the proposed method has superior moving target detection performance in heterogeneous clutter environments.

The remainder of this article is structured as follows. Section II introduces the signal model and the basis of the HPD Riemannian manifold. Section III elaborates on the proposed MIG detector based on the optimal DR mapping of the KLD and its processing framework. Section IV presents the performance analysis and experimental results of the proposed method. Section V gives the conclusions of this article.

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II. Signal Model and Preliminaries on Riemannian Manifold of HPD Matrices

A. Airborne Multichannel SAR Signal Model and Detection Problem Modeling

In this article, we consider multichannel SAR systems with a side-looking uniform linear array, as shown in Fig. 1. The airborne radar system contains N channels distributed along the trajectory, with channel-1 transmitting and channels 1–N receiving simultaneously. The distance between channel-n and channel-1 is $d_n(d_1 = 0)$. The uniform linear array satisfies $d_n = (n-1)d$, with d being the channel spacing. The radar system is moving at a uniform speed at altitude H with a velocity of V. ϕ is the radar down-view angle. The x-axis and y-axis are the direction along the trajectory (azimuthal direction) and the direction perpendicular to the trajectory (range direction), respectively. Let the moving target located in the center of the radar beam at the moment be t_{ac}, while the position of the moving target at this moment be $(x_0, y_0, 0)$. Note the parameters of the moving target as $\boldsymbol{\vartheta}_{cv} = (x_0, y_0, v_x, v_y)$. The synthetic aperture time is $T_{a,cv} = L_{\text{syn}}/(V - v_x)$. In SAR preprocessing, the range compression is performed first for raw echoes. To ensure that the stationary clutter of different channels has consistent amplitude–phase characteristics, channel registration is required after range compression [37]. Finally, multichannel SAR image domain signal is obtained after azimuth compression using the principle of stationary phase (POSP) [38]

$$\begin{aligned}\boldsymbol{s}_{\text{img},cv}\left(\hat{t}, t_m, \boldsymbol{\vartheta}_{cv}\right) = A_{cv}\text{sinc}\left[B_r\left(\hat{t} - \frac{2R_B}{c}\right)\right] \\ \cdot \text{rect}\left(\frac{t_m - t_{ac} + \frac{f_{d,cv}}{K_c}}{B_{cv}/\Delta K}\right) \\ \exp\left[j\pi\Delta K\left(t_m - t_{ac} + \frac{f_{d,cv}}{K_{cv}}\right)^2\right]\Delta_{\text{ATI},cv}\left(\boldsymbol{\vartheta}_{cv}\right)\end{aligned} \tag{1}$$

where A_{cv} represents the scattering intensity of the moving target, λ is the radar wavelength, c is the propagation speed of electromagnetic wave in vacuum, and B_r is the bandwidth. $f_{d,cv} = -2v_r/\lambda$ and $K_{cv} = -2(V - v_x)^2/\lambda R_B$ are the Doppler center frequency and Doppler modulation frequency, respectively. $v_r = v_y \sin\phi$ is the radial velocity of the moving target. In (1), ΔK satisfies $1/\Delta K = 1/K_{cv} - 1/K_c$, where $K_c = -2V^2/\lambda R_B$ is the Doppler modulation frequency of the stationary target. $\Delta_{\text{ATI},cv}(\boldsymbol{\vartheta}_{cv})$ is the multichannel interferometric phase vector of moving target with the expression

$$\begin{aligned}&\Delta_{\text{ATI},cv}\left(\boldsymbol{\vartheta}_{cv}\right) \\ &= \left[1, \exp\left(j\frac{2\ \pi\ d_2\ v_r}{\lambda V}\right), \ldots, \exp\left(j\frac{2\ \pi\ d_N\ v_r}{\lambda V}\right)\right].\end{aligned} \tag{2}$$

It can be seen that the Doppler center of moving target deviates from the zero frequency due to the effect of radial velocity, resulting in an azimuthal shift of the moving target signal in the SAR image domain toward the center by $\Delta t_m = -f_{d,cv}/K_c = -v_r R_B/V^2$, which corresponds to an azimuthal positional shift of the target $\Delta x = -v_r R_B/V$. In addition, the azimuthal compression using the Doppler parameters of a stationary target is mismatched with the Doppler phase history of moving target, causing the imaging result to be defocused, in which case the moving target amplitude response is the multiplication of the range direction sinc(·) and the azimuthal direction rect(·), and the width of the azimuthal magnitude response is $\Delta T = B_{cv}/\Delta K$. Defocus of SAR images of moving targets reduces the target energy, and detecting moving targets in the background of stationary ground clutter is a challenging problem.

In multichannel SAR image domain processing, let the complex-valued sequence of a certain pixel be $\boldsymbol{z}_k = [z_1, z_2, \ldots, z_N]^T$, N is the number of channels, and given the CUT $\boldsymbol{z}$ (the pixel index is omitted for the convenience of derivation), the moving target detection problem can be modeled as a binary hypothesis testing problem

$$\begin{cases}\mathcal{H}_0 : \boldsymbol{z} = \boldsymbol{c} + \boldsymbol{n} \\ \mathcal{H}_1 : \boldsymbol{z} = \alpha\boldsymbol{s} + \boldsymbol{c} + \boldsymbol{n}\end{cases} \tag{3}$$

where $\mathcal{H}_0$ and $\mathcal{H}_1$ denote the absence and presence of target in the CUT, respectively, and α denotes the complex amplitude of the moving target. $\boldsymbol{s}$, $\boldsymbol{c}$, and $\boldsymbol{n}$ denote the multichannel signal vectors corresponding to moving target, clutter, and noise, respectively. $\boldsymbol{s} = \Delta_{\text{ATI},cv}(\boldsymbol{\vartheta}_{cv})$. The optimal adaptive filter for STAP based on linearly constrained minimum variance (LCMV) is [39]

$$\boldsymbol{w}_{\text{opt}} = \mu\boldsymbol{R}^{-1}\boldsymbol{s} \tag{4}$$

where $\mu = 1/(\boldsymbol{s}^H\boldsymbol{R}^{-1}\boldsymbol{s})^{1/2}$, and $\boldsymbol{R}$ is the CCM of CUT, which is estimated by the training samples that are IID with the CUT. Suppose there are K training samples $\{\boldsymbol{z_1}, \boldsymbol{z_1}, \ldots, \boldsymbol{z_K}\}$ in the scene for estimating the CCM, then the SMI estimator is

$$\hat{\boldsymbol{R}}_{\text{SMI}} = \frac{1}{K}\sum_{k=1}^{K}\boldsymbol{z}_k\boldsymbol{z}_k^H. \tag{5}$$

The output of the optimal filter defined by (4) can also be obtained through GLRT derived under the assumption of complex Gaussian clutter [9]

$$t_{\text{STAP}} = \left\|\boldsymbol{w}_{\boldsymbol{opt}}^H\boldsymbol{z}\right\|^2 = \frac{\left|\boldsymbol{s}^H\boldsymbol{R}^{-1}\boldsymbol{z}\right|^2}{\boldsymbol{s}^H\boldsymbol{R}^{-1}\boldsymbol{s}}. \tag{6}$$

Clutter suppression can be achieved by processing each CUT in the multichannel SAR image domain through (6).

B. Multichannel SAR Signal Representation and Geometric CCM Estimation Based on HPD Manifold

To facilitate the representation and processing of the detection problem, the covariance matrix, as a second-order statistic, is often used to represent the statistical characteristic of the observed signal vector. Therefore, we aim to obtain the second-order statistical characteristics (covariance matrices) $\{\boldsymbol{R}_1, \boldsymbol{R}_2, \ldots, \boldsymbol{R}_K\}$ of each sample data $\{\boldsymbol{z_1}, \boldsymbol{z_2}, \ldots, \boldsymbol{z_K}\}$ in the scene. A natural idea is to take it as $\boldsymbol{z}_k\boldsymbol{z}_k^H$. The CCM estimator can be regarded as a function of the matrices in the training dataset, $\hat{\boldsymbol{R}} = f(\boldsymbol{R}_1, \boldsymbol{R}_2, \ldots, \boldsymbol{R}_K)$. The SMI estimator can be considered as the arithmetic mean of the sample covariance

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matrices in the Euclidean space, which is equivalent to the solution of the following optimization problem:

$$\mathcal{P}_1: \begin{cases} \hat{\boldsymbol{R}} = \arg\min\limits_{R>0} \dfrac{1}{K}\sum\limits_{k=1}^{K} d_E^2(\boldsymbol{R}, \boldsymbol{R}_k) \\ \text{s.t.} \quad \hat{\boldsymbol{R}} \in \mathbb{C}^{N\times N} \end{cases} \tag{7}$$

where $\boldsymbol{R}_k$ is taken as $\boldsymbol{z}_k\boldsymbol{z}_k^{\boldsymbol{H}}$ of the sample data, and $d_E^2(\boldsymbol{R}, \boldsymbol{R}_k) = \|\boldsymbol{R}_k - \boldsymbol{R}\|_F^2 = \text{tr}[(\boldsymbol{R}_k - \boldsymbol{R})^H(\boldsymbol{R}_k - \boldsymbol{R})]$ represents the distance between two matrices in the Euclidean space. It should be noted that using the Euclidean distance for measurement only considers the power characteristics of the data and ignores the nonlinear structure of the signal itself. At the same time, the sample covariance $\boldsymbol{z}_k\boldsymbol{z}_k^{\boldsymbol{H}}$ taken at this time is a matrix of rank 1, while the covariance matrix should satisfy the Hermitian positive-definite constraint. To fully consider the nonlinear geometric structure of the signal, we consider solving for the approximate HPD matrix of $\boldsymbol{z}_k\boldsymbol{z}_k^{\boldsymbol{H}}$ through the following constrained optimization problem:

$$\mathcal{P}_{\text{HPD}}: \begin{cases} \min\limits_{\boldsymbol{R}_k} & \left\|\boldsymbol{z}_k\boldsymbol{z}_k^{\boldsymbol{H}} - \boldsymbol{R}_k\right\|^2 \\ \text{s.t.} & \boldsymbol{R}_k \succeq \boldsymbol{I}. \end{cases} \tag{8}$$

The solution to the optimization problem defined in (8) is [40]

$$\boldsymbol{R}_k = \boldsymbol{U}_k\boldsymbol{\Lambda}_k\boldsymbol{U}_k^H \tag{9}$$

where

$$\begin{aligned} \boldsymbol{\Lambda}_k &= \text{diag}\left([\lambda_k, 1, \ldots, 1]\right) \\ \lambda_k &= \max\left(1, \|\boldsymbol{z}_k\|^2\right). \end{aligned} \tag{10}$$

The covariance matrix $\boldsymbol{R}_k$ defined for each sample data $\boldsymbol{z}_k$ through (9) can be expressed as a feature mapping $\boldsymbol{R}_k = \varphi(\boldsymbol{z}_k)$. The feature mapping $\varphi(\cdot)$ transforms an N-dimensional complex vector $\boldsymbol{z}$ into an $N \times N$ HPD matrix R. The set of all $N \times N$ HPD matrices constitutes the feature space, that is, $\mathcal{M}_{++}^N = \{\boldsymbol{R} \in \mathbb{C}^{N\times N} | \boldsymbol{R} = \boldsymbol{R}^H, \boldsymbol{R} \succ 0\}$. In differential geometry theory, the feature space is also called the feature manifold. On this basis, a Riemannian metric $\boldsymbol{G}(\cdot)$ is further defined. At this time, the feature manifold becomes an HPD Riemannian manifold. The inner product of tangent vectors $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ at a point $\boldsymbol{y}$ on the feature manifold is defined as $\langle \boldsymbol{y}_1, \boldsymbol{y}_2\rangle_{\boldsymbol{y}} \triangleq \boldsymbol{y}_1^T\boldsymbol{G}(\boldsymbol{y})\boldsymbol{y}_2$. The geometric distance between two points $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ on the manifold $\mathcal{M}_{++}^N$ is defined as the length of the shortest curve connecting the two points, that is,

$$\begin{aligned} d(\boldsymbol{R}_1, \boldsymbol{R}_2) &\triangleq \min_{\gamma\in\overline{\boldsymbol{R}_1\boldsymbol{R}_2}} \|\gamma\| \\ &= \min_{\gamma\in\overline{\boldsymbol{R}_1\boldsymbol{R}_2}} \int_0^1 \sqrt{\left\langle \frac{d\gamma}{dt}, \frac{d\gamma}{dt}\right\rangle_{\gamma(t)}} dt \end{aligned} \tag{11}$$

where $\overline{\boldsymbol{R}_1\boldsymbol{R}_2}$ is the set of curves on the manifold $\mathcal{M}_{++}^N$ that satisfy $\gamma(0) = \boldsymbol{R}_1$ and $\gamma(1) = \boldsymbol{R}_2$. In statistical signal processing problems, the metric tensor $\boldsymbol{G}(\cdot)$ is usually defined as the Fisher information matrix. At this time, the AIRM can be derived [41]

$$d_{\boldsymbol{R}}(\boldsymbol{R}_1, \boldsymbol{R}_2) = \sqrt{\text{tr}\left(\log^2\left(\boldsymbol{R}_1^{-\frac{1}{2}}\boldsymbol{R}_2\boldsymbol{R}_1^{-\frac{1}{2}}\right)\right)}$$

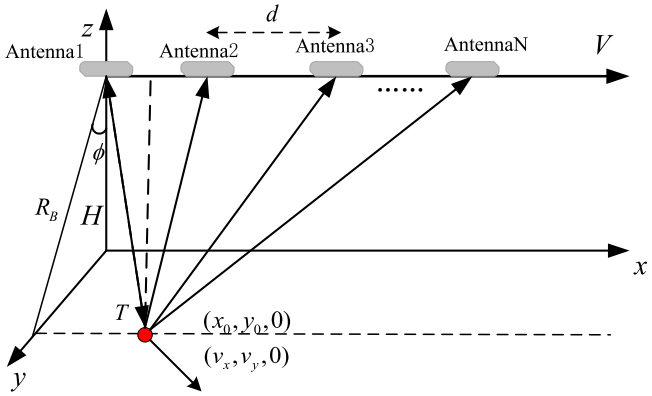


Fig. 1. Geometric model of the multichannel SAR system.

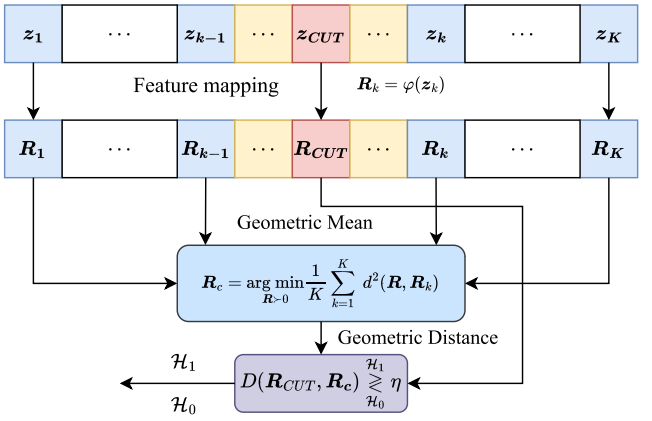


Fig. 2. Radar target adaptive detection framework based on IG.

$$= \left\|\log\left(\boldsymbol{R}_1^{-\frac{1}{2}}\boldsymbol{R}_2\boldsymbol{R}_1^{-\frac{1}{2}}\right)\right\|_F. \tag{12}$$

In addition to AIRM, commonly used geometric metrics also include the LEM [21] and information divergences, such as the KLD [28]. Since (9) transforms each training sample into a corresponding HPD matrix, thus corresponding to different points on the HPD manifold, the problem of estimating the CCM can be transformed into the problem of estimating the geometric mean of the set of sample points on the HPD manifold. The geometric mean of the HPD matrices is used as the covariance matrix estimator, that is,

$$\boldsymbol{R}_c = \arg\min_{\boldsymbol{R}>0} \frac{1}{K}\sum_{k=1}^{K} d^2(\boldsymbol{R}, \boldsymbol{R}_k) \tag{13}$$

where $d^2(\cdot)$ is the selected geometric metric. Existing literatures [40], [42], [43] show that geometric covariance estimation fully considers the nonlinear structure of the signal and has higher estimation accuracy. Replacing the original SMI CCM estimator in the STAP defined by (6) with (13) can effectively improve the clutter suppression performance. In this article, this method is denoted as STAP (geometry).

C. Heterogeneous Clutter Statistical Model and KLD-MIG

The STAP filter defined by (6) is the optimal match under the assumption of complex Gaussian clutter (homogeneous clutter). However, in practical scenarios, clutter often does

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not satisfy the assumption of complex Gaussian distribution. Therefore, the performance of using (6) for clutter suppression is not optimal. Considering the physical mechanism of clutter scattering, a more common statistical model for heterogeneous clutter is the compound Gaussian model, which models the clutter as the product of a fast-varying speckle component and a slow-varying texture component

$$\boldsymbol{c} = \sqrt{\tau} \cdot \boldsymbol{g} = \Delta \cdot \boldsymbol{g} \tag{14}$$

where the speckle component $\boldsymbol{g}$ follows a zero-mean circular complex Gaussian distribution, and the texture component τ is a random variable. When τ is a constant, it represents homogeneous clutter. For heterogeneous scenarios, many studies have investigated statistical models suitable for τ, including Gamma texture [44], inverse Gamma texture [45], inverse Gaussian texture [46], etc. In this article, the inverse Gamma texture described in [45] is used to represent the heterogeneity, and probability density function of Δ is

$$f_{\Delta}(\delta) = \frac{2(\nu-1)^{\nu}}{\Gamma(\nu)} \delta^{-(2\nu+1)} \exp\left(-\frac{\nu-1}{\delta^2}\right) \tag{15}$$

where ν is the clutter texture parameter. The larger the ν, the more homogeneous the clutter scene. It is worth noting that although the optimal detector in a statistical sense can be constructed for a specific texture distribution, due to the difficulty in accurately statistically modeling clutter in com plex backgrounds, when the actual situation does not match the assumed model, the performance of detection methods based on statistical distribution model will decline and often cannot meet the requirements of practical applications. In recent years, target detection techniques based on IG have provided a new processing framework for radar adaptive target detection. They construct a feature manifold by extracting the differential features between target and clutter and use the feature difference metric function on the feature manifold to characterize the data differences to distinguish between the two. Since such methods do not rely on specific probability distribution assumptions of the clutter model, introducing them into radar adaptive detection is expected to improve the performance of traditional statistical detectors. The radar target adaptive detection framework based on IG is shown in Fig. 2. For the research of this article, the extracted features are the second-order statistical characteristics (covariance matrices) of the data, and the feature differences are characterized by KLD. The KLD-MIG can be expressed as follows [28]:

$$\mathcal{D}_{\mathrm{KL}}(\boldsymbol{R}_{\mathrm{CUT}}, \boldsymbol{R}_c) = \mathrm{tr}\left(\boldsymbol{R}_c^{-1}\boldsymbol{R}_{\mathrm{CUT}}\right) - \log\det\left(\boldsymbol{R}_c^{-1}\boldsymbol{R}_{\mathrm{CUT}}\right) - N \tag{16}$$

where $\boldsymbol{R}_{\mathrm{CUT}}$ is the covariance matrix of the data unit to be detected, and $\boldsymbol{R}_c$ is the CCM estimated using the geometric center of the sample covariances of the training units.

III. Proposed Method via KLD-Based Optimal DR

A. Properties of Sample Distribution Characteristics on HPD Riemannian Manifold

According to (9), the sample data z_k of each pixel unit correspond to different points on the HPD Riemannian manifold through the feature mapping $\boldsymbol{R}_k = \varphi(z_k)$. The MIG detector essentially differentiates between samples under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses using the geometric distance of these samples on the HPD manifold. Therefore, analyzing the distribution characteristics of the target and clutter samples on the HPD manifold can intuitively describe the distinguishability of the samples under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses. Although manifold learning algorithms such as Isomap [33], LLE [34], and LE [35] can unfold a single continuous low-dimensional manifold, the t-distributed stochastic neighbor embedding (t-SNE) algorithm [47] has more advantages if one wants to accurately visualize the similarity relationships between samples. Therefore, we use the t-SNE algorithm below to analyze the distribution characteristics of the clutter and target sample data on the HPD manifold under the $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses. The simulation parameters are given in Table I.

TABLE I
System Simulation Parameters for Experiments

Parameter	Symbol	Value
Carrier frequency	f_c	9.6GHz
Sampling frequency	f_s	120MHz
Bandwidth	B_r	100MHz
Pulse repetition frequency	PRF	1500Hz
Velocity	V	300m/s
Height	H	6km
Number of channels	N	6
Channel spacing	d	1m
Clutter to noise ratio	CNR	20dB
Clutter texture	ν	5

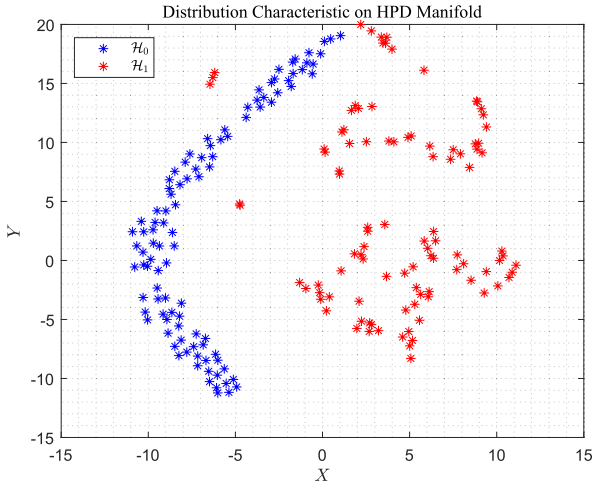
We consider the following three experimental setups: 1) SCR = −5 dB, v_r = 1 m/s; 2) SCR = −8 dB, v_r = 1 m/s; and 3) SCR = −5 dB, v_r = 0.5 m/s. Among them, experimental setups 2) and 3) correspond to the cases of low SCR and slow-moving targets, respectively. Under each experimental setup, we generate 100 groups of data under the $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses. The data under $\mathcal{H}_0$ hypothesis are generated using the clutter statistical model described by (14) and (15), with the clutter texture parameter ν = 5. The data under $\mathcal{H}_1$ hypothesis are obtained by adding the moving target signal vector described by (3) to the data under $\mathcal{H}_0$ hypothesis. The distribution characteristics of the target and clutter under the three experimental setups are shown in Fig. 3.

From the results shown in Fig. 3, it can be seen that for experimental setup 1), the targets and clutter exhibit clustering characteristics on the manifold respectively, showing good distinguishability. For the low SCR scenario 2) and the slow-moving target scenario 3), the distinguishability between the targets and clutter on the manifold is poor, which is not conducive to moving target detection.

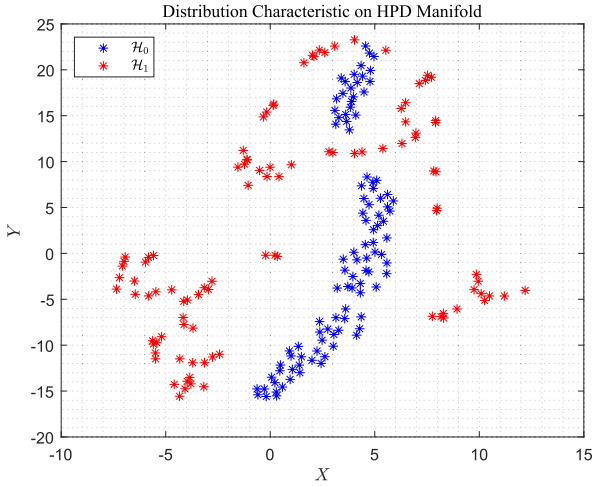
B. Optimal DR Problem via KLD

1) Optimization Problem Modeling and Solution: The MIG detector essentially differentiates data based on the differences in their distributions on the HPD manifold. By projecting the $N \times N$ covariance matrices into $n \times n$ covariance matrices through manifold projection, it is possible to increase the geometric distance of the data on the manifold under the

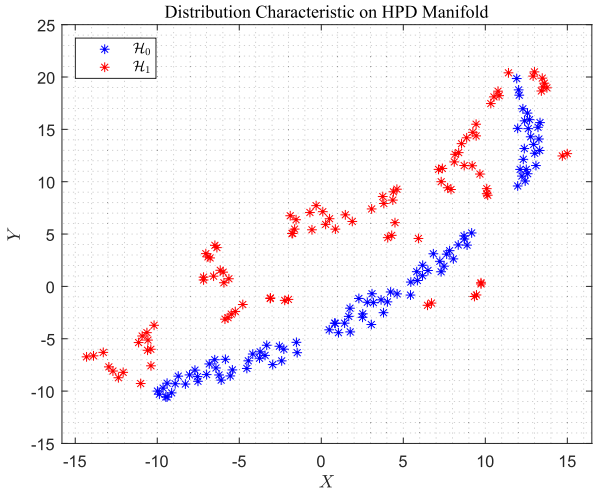
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(a)



(b)



(c)

Fig. 3. Distribution characteristics of the target and clutter sample data on the HPD manifold: (a) SCR = −5 dB and v_r = 1 m/s; (b) SCR = −8 dB and v_r = 1 m/s; and (c) SCR = −5 dB and v_r = 0.5 m/s.

assumptions of $\mathcal{H}_0$ and $\mathcal{H}_1$, making it easier to distinguish and thereby improving the performance of the MIG detector, as shown in Fig. 4.

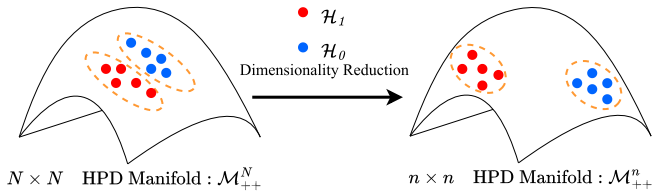


Fig. 4. DR on HPD manifold.

Define the mapping transformation from manifold to manifold.

Definition 1: For the high-dimensional HPD manifold space $\mathcal{M}_{++}^{N}$, a mapping transformation is defined as follows:

$$\begin{aligned} &\mathcal{F}_{\boldsymbol{W}} : \mathcal{M}_{++}^{N} \rightarrow \mathcal{M}_{++}^{n} \\ &\mathcal{F}_{\boldsymbol{W}}(\boldsymbol{R}) = \boldsymbol{W}^{H}\boldsymbol{R}\boldsymbol{W} \end{aligned} \tag{17}$$

where $\boldsymbol{R}$ is an arbitrary point on the HPD manifold $\mathcal{M}_{++}^{N}$, and $\boldsymbol{W} \in \mathbb{C}^{N\times n}$ is a column-full-rank matrix.

After DR mapping, the KLD between the matrices $\boldsymbol{R}_{\mathrm{CUT}}$ and $\boldsymbol{R}_{\boldsymbol{c}}$ becomes

$$\mathcal{D}_{\mathrm{KL}}\left(\boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R}_{\mathrm{CUT}}\boldsymbol{W}, \boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R}_{\boldsymbol{c}}\boldsymbol{W}\right). \tag{18}$$

For a fixed parameter n, DR mapping is determined by the column-full-rank matrix $\boldsymbol{W}$. Consider the DR problem of maximizing the KLD under the two hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$

$$\mathcal{P}_{\mathrm{DR}} : \begin{cases} \max\limits_{\boldsymbol{W}} & \mathcal{D}_{\mathrm{KL}}\left(\boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R}_{\boldsymbol{1}}\boldsymbol{W}, \boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R}_{\boldsymbol{0}}\boldsymbol{W}\right) \\ \text{s.t.} & \boldsymbol{W} \in \mathbb{C}^{N\times n},\ \mathrm{rank}(\boldsymbol{W}) = n \end{cases} \tag{19}$$

where $\boldsymbol{R}_{\boldsymbol{1}} = \boldsymbol{R}_{\boldsymbol{c}} + \boldsymbol{R}_{\boldsymbol{t}}$ and $\boldsymbol{R}_{\boldsymbol{0}} = \boldsymbol{R}_{\boldsymbol{c}}$. $\boldsymbol{R}_{\boldsymbol{t}}$ is the target covariance matrix. Since the target is represented as the product of the scattering coefficient and the steering vector $\alpha_t \boldsymbol{s}_t$, the form of the target covariance matrix is $\boldsymbol{R}_{\boldsymbol{t}} = \alpha_t \alpha_t^{H} \boldsymbol{s}_t \boldsymbol{s}_t^{H}$. In practical processing, the complex amplitude of the target and the target steering vector are unknown, and it is necessary to estimate them using the CUT $\boldsymbol{z}_{\mathrm{CUT}}$. The target steering vector can be estimated by the AMF test statistic

$$\hat{\boldsymbol{s}}_t = \arg\max_{\boldsymbol{s}\left(v_r^{(m)}\right)} \frac{\left|\boldsymbol{s}\left(v_r^{(m)}\right)^{H} \boldsymbol{R}_{\boldsymbol{c}}^{-\boldsymbol{1}} \boldsymbol{z}_{\mathrm{CUT}}\right|^{2}}{\boldsymbol{s}\left(v_r^{(m)}\right)^{H} \boldsymbol{R}_{\boldsymbol{c}}^{-\boldsymbol{1}} \boldsymbol{s}\left(v_r^{(m)}\right)}. \tag{20}$$

Furthermore, the minimum-variance unbiased estimate (MVUE) of the target complex amplitude is obtained

$$\hat{\alpha}_t = \frac{\hat{\boldsymbol{s}}_t^{H} \boldsymbol{R}_{\boldsymbol{c}}^{-\boldsymbol{1}} \boldsymbol{z}_{\mathrm{CUT}}}{\hat{\boldsymbol{s}}_t^{H} \boldsymbol{R}_{\boldsymbol{c}}^{-\boldsymbol{1}} \hat{\boldsymbol{s}}_t}. \tag{21}$$

Now we obtain the estimate of the target covariance matrix $\boldsymbol{R}_t$, and further obtain the covariance matrix $\boldsymbol{R}_1$ under $\mathcal{H}_1$ hypothesis. Next, we solve the optimization problem described by (19). Noting that the KLD is affine invariant, we first consider making a variable substitution $\boldsymbol{Y} = \boldsymbol{R}_0^{1/2}\boldsymbol{W}$ for prewhitening processing. Since $\boldsymbol{R}_0^{1/2}$ is a full-rank matrix of size $N \times N$ and $\mathrm{rank}(\boldsymbol{W}) = n$, $\boldsymbol{Y}$ is also a column-full-rank matrix of size $N \times n$, and $\mathrm{rank}(\boldsymbol{Y}) = n$. Consider the QR decomposition of $\boldsymbol{Y}$, $\boldsymbol{Y} = \boldsymbol{Q}\boldsymbol{R}$ ($\boldsymbol{Q}^{H}\boldsymbol{Q} = \boldsymbol{I}_n$, $\mathrm{rank}(\boldsymbol{R}) = n$), and

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then substitute $\boldsymbol{W} = \boldsymbol{R}_0^{-1/2}\boldsymbol{Y}$ into the terms in the objective function. Thus, we have

$$\begin{aligned}\boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R_0}\boldsymbol{W} &= \left(\boldsymbol{R}_0^{-1/2}\boldsymbol{Y}\right)^{H}\boldsymbol{R}_0\left(\boldsymbol{R}_0^{-1/2}\boldsymbol{Y}\right)\\ &= \boldsymbol{Y}^{H}\boldsymbol{R}_0^{-1/2}\boldsymbol{R}_0\boldsymbol{R}_0^{-1/2}\boldsymbol{Y}\\ &= \boldsymbol{Y}^{H}\boldsymbol{Y}\\ &= \boldsymbol{R}^{H}\boldsymbol{R} \end{aligned} \tag{22}$$

$$\begin{aligned}\boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R_1}\boldsymbol{W} &= \left(\boldsymbol{R}_0^{-1/2}\boldsymbol{Y}\right)^{H}\boldsymbol{R_1}\left(\boldsymbol{R}_0^{-1/2}\boldsymbol{Y}\right)\\ &= \boldsymbol{Y}^{H}\left(\boldsymbol{R}_0^{-1/2}\boldsymbol{R_1}\boldsymbol{R}_0^{-1/2}\right)\boldsymbol{Y}\\ &= \boldsymbol{R}^{H}\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\boldsymbol{R} \end{aligned} \tag{23}$$

where $\tilde{\boldsymbol{R}}_1 = \boldsymbol{R}_0^{-1/2}\boldsymbol{R_1}\boldsymbol{R}_0^{-1/2}$ is the covariance matrix obtained by whitening $\boldsymbol{R}_1$ with respect to $\boldsymbol{R}_0$. At this time, we have

$$\begin{aligned}&\left(\boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R_0}\boldsymbol{W}\right)^{-1}\boldsymbol{W}^{\boldsymbol{H}}\boldsymbol{R_1}\boldsymbol{W}\\ &= \left(\boldsymbol{R}^{H}\boldsymbol{R}\right)^{-1}\left(\boldsymbol{R}^{H}\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\boldsymbol{R}\right)\\ &= \boldsymbol{R}^{-1}\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\right)\boldsymbol{R}. \end{aligned} \tag{24}$$

Using the properties of the trace and the determinant

$$\begin{aligned}&\operatorname{tr}\left[\boldsymbol{R}^{-1}\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\right)\boldsymbol{R}\right]\\ &= \operatorname{tr}\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\right)\det\left[\boldsymbol{R}^{-1}\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\right)\boldsymbol{R}\right] \end{aligned} \tag{25}$$

$$\begin{aligned}&= \det\left(\boldsymbol{R}^{-1}\right)\det\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\right)\det\left(\boldsymbol{R}\right)\\ &= \det\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}}_1\boldsymbol{Q}\right). \end{aligned} \tag{26}$$

The objective function can be written in a form that only depends on the orthonormal basis $\boldsymbol{Q}$

$$g\left(\boldsymbol{Q}\right) = \operatorname{tr}\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}_1}\boldsymbol{Q}\right) - \log\det\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}_1}\boldsymbol{Q}\right) - n. \tag{27}$$

Since the objective function does not depend on $\boldsymbol{R}$, the optimization problem is transformed into finding the optimal $\boldsymbol{Q}$ on the Grassmann manifold $\mathcal{G}(n, N)$

$$\mathcal{P}'_{\mathrm{DR}}: \begin{cases}\max\limits_{\boldsymbol{Q}} & g\left(\boldsymbol{Q}\right) = \operatorname{tr}\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}_1}\boldsymbol{Q}\right) - \log\det\left(\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}_1}\boldsymbol{Q}\right) - n\\ \text{s.t.} & \boldsymbol{Q}\in\mathbb{C}^{N\times n}, \boldsymbol{Q}^{H}\boldsymbol{Q} = \boldsymbol{I}_n. \end{cases} \tag{28}$$

The optimization problem defined by (28) can be solved by the Riemannian conjugate gradient method [36]. However, this method requires iterative solutions on the Grassmann manifold and has a large computational cost. Since the processing of the multichannel SAR image domain is carried out pixel-by-pixel, this algorithm is not very suitable for the huge data load of the SAR system. To reduce the computational complexity of the optimization problem, we attempt to transform the optimization problem (28) into an eigenvalue problem of the HPD matrix using the equivalent representation of the KLD of the HPD matrix. First, we consider the equivalent representation of the objective function. Suppose the eigenvalues of $\tilde{\boldsymbol{R}}_1$ are arranged in descending order as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N > 0$, and the eigenvalues of $\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}_1}\boldsymbol{Q}$ are arranged in descending order as $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n > 0$. The objective function can be expressed as $g(\boldsymbol{Q}) = \sum_{j=1}^{n}\mu_j - \log\prod_{j=1}^{n}\mu_j - n$. Define function

$$\psi\left(\mu\right) = \mu - \log\mu - 1. \tag{29}$$

For the convenience of subsequent analysis, we first present Proposition 1.

Proposition 1: For a Hermitian positive definite matrix $\tilde{\boldsymbol{R}}_1$, assume that its eigenvalues are arranged in descending order as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N > 0$. Then, for $\forall \boldsymbol{Q} \in \mathbb{C}^{N\times n}$ with $\boldsymbol{Q}^{H}\boldsymbol{Q} = \boldsymbol{I}_n$, $\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}_1}\boldsymbol{Q}$ is also a Hermitian positive definite matrix. Moreover, if its eigenvalues are arranged in descending order as $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n > 0$, the eigenvalues should satisfy the constraints: $\lambda_{N-n+k} \leq \mu_k \leq \lambda_k$, where $k = 1, \ldots, n$.

Proof: See the Appendix.□

Since $d\psi(\mu)/d\mu = 1 - 1/\mu$, this function is monotonically decreasing on the interval $(0, 1)$ and monotonically increasing on the interval $(1, +\infty)$. According to Proposition 1, the eigenvalues should satisfy the constraints: $\lambda_{N-n+k} \leq \mu_k \leq \lambda_k$, for $k = 1, \ldots, n$. Therefore, the optimal eigenvalues should be obtained at the endpoints

$$\mu_k^* = \underset{\{\lambda_k, \lambda_{N-n+k}\}}{\arg\max}\left(\mu_k - \log\mu_k - 1\right). \tag{30}$$

Next, we need to construct an orthogonal matrix $\boldsymbol{Q}$ such that the eigenvalues of $\boldsymbol{Q}^{H}\tilde{\boldsymbol{R}_1}\boldsymbol{Q}$ are exactly $\mu_1^*, \mu_2^*, \ldots, \mu_N^*$. This is a variant of the inverse eigenvalue problem, that is, the inverse eigenvalue problem of the projection of an HPD matrix into a subspace. Generally, it is complex to explicitly construct $\boldsymbol{Q}$, but in the context of the optimization problem $\mathcal{P}'_{\mathrm{DR}}$, this is greatly simplified. As can be seen from (30), the optimal eigenvalue set $\{\mu_1^*, \ldots, \mu_n^*\}$ is actually a subset of the eigenvalue set $\{\lambda_1, \ldots, \lambda_N\}$. Specifically, for each $k \in \{1, \ldots, n\}$, the optimal μ_k^* is either λ_k or λ_{N-n+k}. Let $\mathrm{I}^* = \{i_1, \ldots, i_n\}$ be the set of indices of these n selected optimal eigenvalues in the eigenvalue list $\lambda_1, \ldots, \lambda_N$. At this time, $\{\mu_1^*, \ldots, \mu_n^*\} = \{\lambda_{i_1}, \lambda_{i_2}, \ldots, \lambda_{i_n}\}$, and the construction of the matrix $\boldsymbol{Q}$ can be achieved by selecting the eigenvectors corresponding to the index set. Specifically, we first calculate the eigenvalue decomposition of $\tilde{\boldsymbol{R}}_1$ $\tilde{\boldsymbol{R}}_1 = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^{H}$, where $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_N)$ and $\lambda_1 \geq \cdots \geq \lambda_N > 0$, and $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_N]$ is the corresponding unitary eigenvector matrix ($\boldsymbol{U}^{H}\boldsymbol{U} = \boldsymbol{I}_N$). Then, we determine the optimal eigenvalue set: For $k = 1, \ldots, n$, compare $\psi(\lambda_k) = \lambda_k - \log\lambda_k - 1$ and $\psi(\lambda_{N-n+k}) = \lambda_{N-n+k} - \log\lambda_{N-n+k} - 1$. If $\psi(\lambda_k) \geq \psi(\lambda_{N-n+k})$, then select $\mu_k^* = \lambda_k$, and the corresponding index is $i_k = k$, else if $\psi(\lambda_k) < \psi(\lambda_{N-n+k})$, then select $\mu_k^* = \lambda_{N-n+k}$, and the corresponding index is $i_k = N-n+k$. Finally, we construct $\boldsymbol{Q}$. Select the eigenvectors corresponding to the index set I^*. Construct the matrix $\boldsymbol{Q}$ such that its column vectors are $\{\boldsymbol{u}_i | i \in \mathrm{I}^*\}$. That is, $\boldsymbol{Q} = [\boldsymbol{u}_{i_1}, \boldsymbol{u}_{i_2}, \ldots, \boldsymbol{u}_{i_n}]$. It has been proven in (28) that the KLD after DR is independent of $\boldsymbol{R}$ (only requiring $\boldsymbol{R}$ to be invertible), so we can choose $\boldsymbol{R} = \boldsymbol{I}_n$. So far, we have obtained the optimal DR matrix $\boldsymbol{W}_{\mathrm{opt}} = \boldsymbol{R}_0^{-1/2}\boldsymbol{Q}$.

2) Process of the Proposed Algorithm and Computational Complexity Analysis: The process of constructing the optimal DR matrix based on KLD can be summarized as follows.

Based on the process of Algorithm 1, we analyze the computational complexity of the algorithm. Assume that the Gaussian elimination method is used for matrix inversion and the singular value decomposition (SVD) is used for square-root operations. For a matrix of dimension $N \times N$, the complexity of such operations is $\mathcal{O}(N^3)$. The computational complexity of solving $\boldsymbol{R}_0^{-1/2}$ and multiplication in Step 2 is

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Algorithm 1 Optimal Projection Matrix Construction

Require: HPD matrices $\boldsymbol{R_1}, \boldsymbol{R_0} \in \mathbb{C}^{N\times N}$; Reduced dimension n $(1 \le n < N)$.

Ensure: Optimal projection matrix $\boldsymbol{W_{opt}} \in \mathbb{C}^{N\times n}$.

1: Initialize: Index set $I^* \leftarrow \emptyset$.
2: Compute whitened matrix: $\tilde{\boldsymbol{R}}_1 \leftarrow \boldsymbol{R_0}^{-1/2}\boldsymbol{R_1}\boldsymbol{R_0}^{-1/2}$.
3: Compute eigenvalue decomposition of $\tilde{\boldsymbol{R}}_1$: $\tilde{\boldsymbol{R}}_1 = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^H$, where $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$ with $\lambda_1 \ge \ldots \ge \lambda_N > 0$, and $\boldsymbol{U} = [\boldsymbol{u_1}, \ldots, \boldsymbol{u_N}]$, where $\boldsymbol{U}$ is unitary, and $\boldsymbol{u_k}$ is the kth eigenvector
4: **for** $k = 1$ to n **do**
5: **if** $\psi(\lambda_k) \ge \psi(\lambda_{N-n+k})$ **then**
6: $\mu_k^* \leftarrow \lambda_k$
7: $\boldsymbol{i_k} \leftarrow k$
8: **elseif** $\psi(\lambda_k) < \psi(\lambda_{N-n+k})$
9: $\mu_k^* \leftarrow \lambda_{N-n+k}$
10: $\boldsymbol{i_k} \leftarrow N - n + k$
11: **end if**
12: $I^* \leftarrow I^* \cup \{\boldsymbol{i_k}\}$
13: **end for**
14: Construct matrix $\boldsymbol{Q}$ from selected eigenvectors: $\boldsymbol{Q} \leftarrow [\boldsymbol{u}_{i_1}, \boldsymbol{u}_{i_2}, \ldots, \boldsymbol{u}_{i_n}]$.
15: Compute optimal projection matrix: $\boldsymbol{W_{opt}} \leftarrow \boldsymbol{R_0}^{-1/2}\boldsymbol{Q}$.
16: **return** $\boldsymbol{W_{opt}}$

$\mathcal{O}(N^3)$, the computational complexity of Step 3 is $\mathcal{O}(N^3)$, the computational complexity of Steps 4–14 is $\mathcal{O}(N)$, and the computational complexity of Step 15 is $\mathcal{O}(N^2 n)$. Therefore, the computational complexity of the proposed algorithm is mainly determined by the eigenvalue decomposition, and the overall computational complexity is $\mathcal{O}(N^3)$. The solution using the Riemannian conjugate gradient method on the Grassmann manifold has an SVD operation in each iteration [36], with a computational complexity of $\mathcal{O}(N^3)$. If the number of iterations required for its convergence is N_{iter}, the overall complexity of this algorithm is $\mathcal{O}(N_{\mathrm{iter}}N^3)$. In summary, the computational complexity of the proposed algorithm $\mathcal{O}(N^3)$ is significantly lower than that of the Riemannian conjugate gradient method on the Grassmann manifold $\mathcal{O}(N_{\mathrm{iter}}N^3)$, making its application in multichannel SAR moving target detection possible.

C. Multichannel SAR Moving Target Detection Framework Based on DR KLD-MIG

In summary, the proposed detection framework mainly includes the following steps.

Step1: the multichannel SAR image domain signal is obtained by performing range compression, channel registration, and azimuth compression on the multichannel SAR echo signal.
Step2: select training samples using generalized inner product (GIP) criterion.
Step3: define the covariance matrix $\boldsymbol{R}_k$ for each sample data z_k through (9) to get the set of covariance matrices $\{\boldsymbol{R}_1, \boldsymbol{R}_2, \ldots, \boldsymbol{R}_K\}$.
Step4: estimate the CCM using the geometric mean of the training samples:

$$\boldsymbol{R}_c = \arg\min_{\boldsymbol{R}>0} (1/K) \sum_{k=1}^{K} d^2(\boldsymbol{R}, \boldsymbol{R}_k).$$

Step5: obtain the estimate of the target covariance matrix using (20) and (21).
Step6: obtain the optimal DR matrix using Algorithm 1.
Step7: perform KLD-MIG after DR: $\mathcal{D}_{\mathrm{KL}}(\boldsymbol{W^H R}_{\mathrm{CUT}}\boldsymbol{W}, \boldsymbol{W^H R_c W})$.
Step8: apply constant false alarm rate (CFAR) detector to the result after KLD-MIG.

Fig. 5 shows the framework of multichannel SAR moving target enhancement detection via DR KLD-MIG.

IV. Experimental Results and Analysis

In this section, the performance of the proposed method will be discussed in detail. The simulation environment uses the Windows 11 operating system, with a 12th-Gen Intel[1] Core[2] i7-12700H processor and 32 GB of system memory. The parameters of the radar system used in the experiment are given in Table I. Section IV-A analyzes the manifold distribution characteristics after data DR mapping. Section IV-B discusses the performance of the proposed method in detail. Sections IV-C and IV-D present the clutter suppression and target detection results of the proposed method under simulated and measured clutter backgrounds, respectively.

A. Properties of Sample Distribution Characteristics After DR

To intuitively illustrate the effectiveness of the DR mapping, this section will present the visualization results of the distribution characteristics of the data on the HPD manifold after DR mapping. We still consider the same three experimental setups as in Section III-A, and with the reduction dimension $n = 3$, the DR mapping matrix is calculated using Algorithm 1 and the data are subjected to DR processing. Then, the t-SNE algorithm is used to visualize the results as shown in Fig. 6.

By comparing Figs. 3(a) with 6(a), Figs. 3(b) with 6(b), and Figs. 3(c) with 6(c), it can be observed that after the DR mapping process, the distinguishability of the data under the two hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$ is further enhanced compared with that on the original HPD manifold. On the original HPD manifold, in the scenarios of low SCR and slow-moving targets, the data under $\mathcal{H}_0$ and $\mathcal{H}_1$ are hardly distinguishable. However, after the DR process, the data under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses in these two situations show obvious clustering characteristics and possess separability. This indicates that the detection performance of the MIG detector is expected to be improved.

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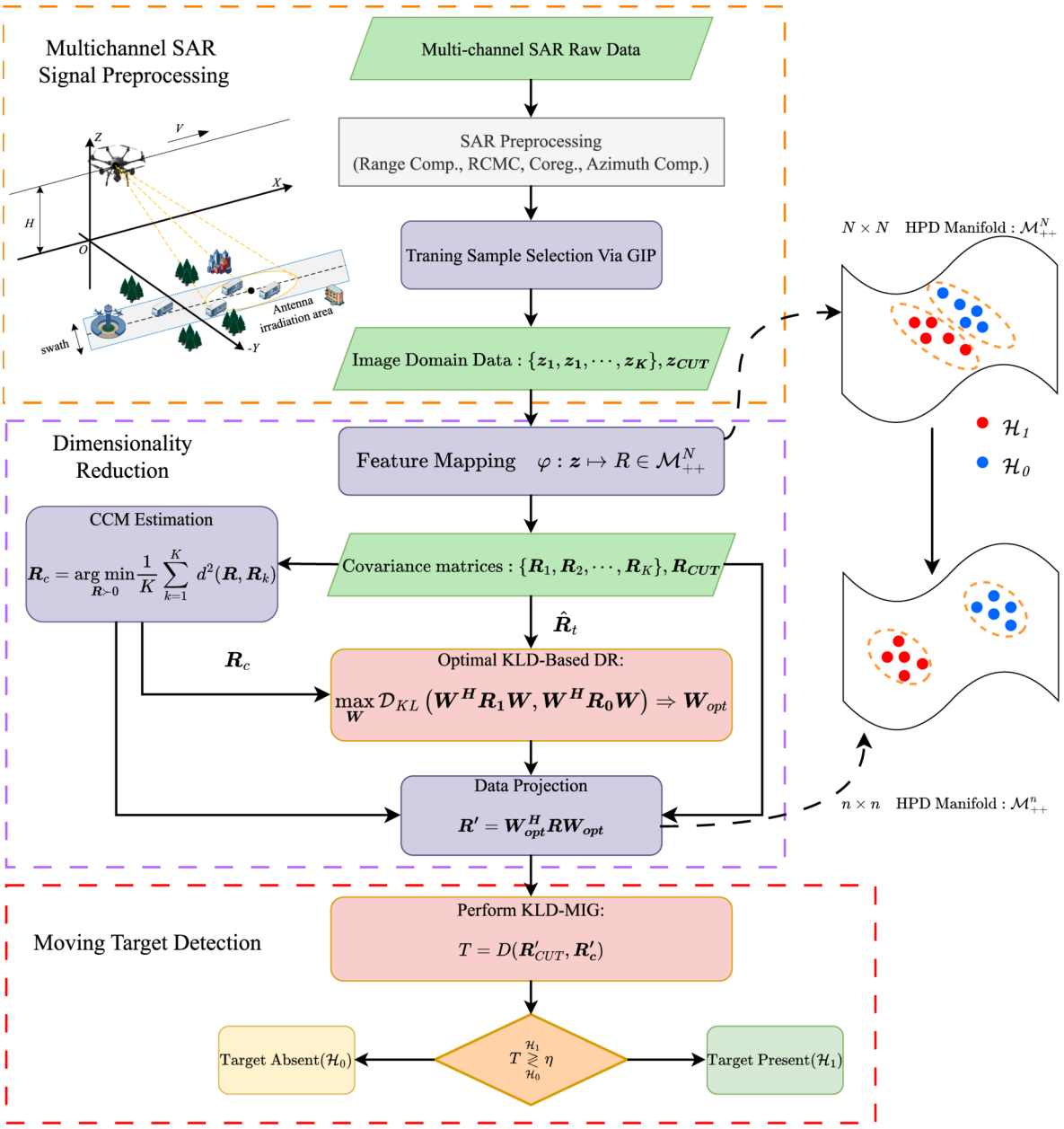


Fig. 5. Multichannel SAR detection framework via DR KLD-MIG.

B. Performance Analysis of Proposed Method

1) Analysis of Data Distinguishability Performance: In this section, we further quantitatively characterize the distinguishability of the data under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses by DR mapping. The clutter texture parameter is still set to $\nu = 5$. First, 100 groups of data under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses are generated, respectively, and the geometric mean of the data under $\mathcal{H}_0$ hypothesis is used to estimate the CCM. Subsequently, the average KLD $\mathcal{D}_{\mathrm{KL}}(\mathcal{H}_1 \| \mathcal{H}_0)$ of the data under the $\mathcal{H}_1$ hypothesis to the CCM and the average KLD $\mathcal{D}_{\mathrm{KL}}(\mathcal{H}_0 \| \mathcal{H}_0)$ of the data under the $\mathcal{H}_0$ hypothesis to the CCM are calculated. The distinguishability of the data under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses can be quantitatively expressed by the scatter ratio

$$\text{Scatter Ratio} = 10 \log_{10} \left(\frac{\mathcal{D}_{\mathrm{KL}}(\mathcal{H}_1 \| \mathcal{H}_0)}{\mathcal{D}_{\mathrm{KL}}(\mathcal{H}_0 \| \mathcal{H}_0)} \right). \quad (31)$$

In the experiment, we first fix the radial velocity at $v_r = 1$ m/s and vary the input SCR to evaluate the change in the scatter ratio with the input SCR. Then, we fix the input SCR at -5 dB and vary the radial velocity to evaluate the change in the scatter ratio with the radial velocity. The evaluation range of the SCR is from -15 to 5 dB, and the evaluation range of the radial velocity is from -3.5 to 3.5 m/s. After 100 independent Monte Carlo experiments, the obtained results are as shown in Fig. 7.

From the experimental results, it can be seen that when the radial velocity is fixed, the scatter ratio of each method gradually increases with the increase in the input SCR. When the input SCR is the same, the scatter ratio is the largest when the dimension is $n = 1$, and the scatter ratio without DR processing is the smallest. When the input SCR is fixed, the scatter ratio of each method gradually increases with the

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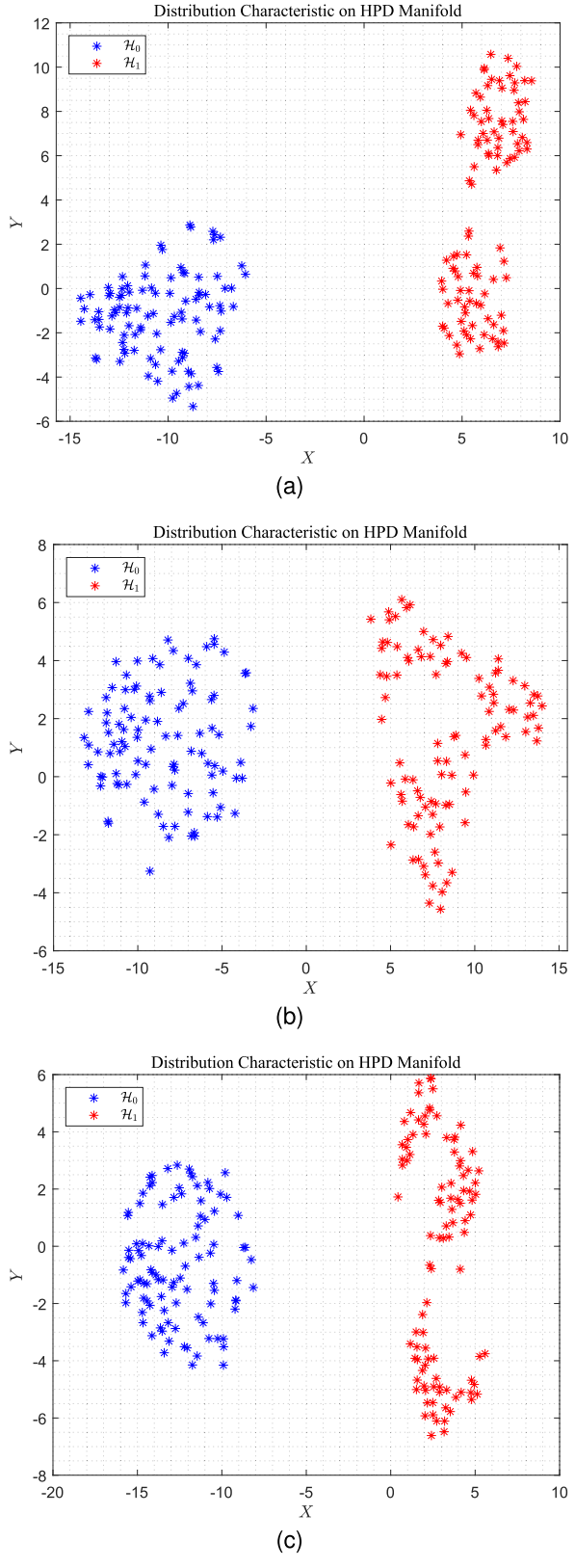


Fig. 6. Distribution characteristics of the target and clutter sample data on the HPD manifold after DR: (a) SCR = −5 dB and v_r = 1 m/s; (b) SCR = −8 dB and v_r = 1 m/s; and (c) SCR = −5 dB and v_r = 0.5 m/s.

increase in the radial velocity. When the dimension is $n = 1$, the notch of the curve is the narrowest, indicating that it has strong distinguishability even when the radial velocity is small.

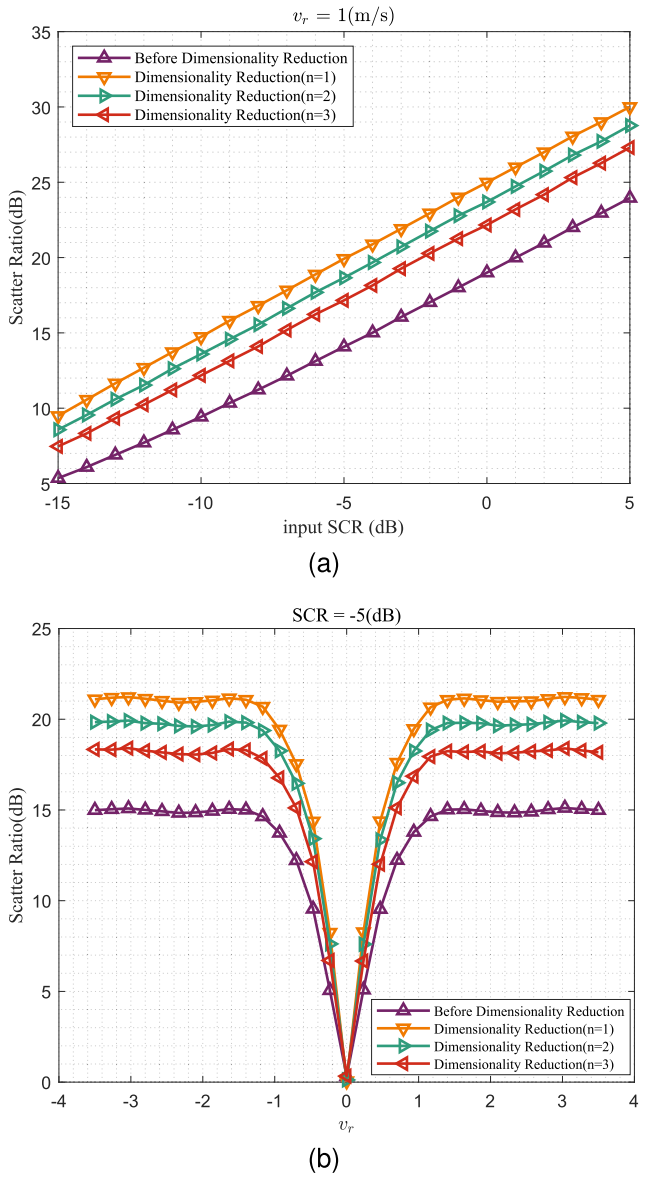


Fig. 7. Distinguishability performance of the proposed method: (a) scatter ratio versus SCR and (b) scatter ratio versus v_r.

While the notch without DR processing is the widest, and the distinguishability is the worst at the same velocity. Thus, it can be concluded that the distinguishability of the data after DR mapping is significantly higher than that of the data on the original HPD manifold, and the lower the DR dimension, the stronger the distinguishability of the data under $\mathcal{H}_0$ and $\mathcal{H}_1$ hypotheses.

2) Analysis of Detection Performance: In this section, the detection performance of the proposed method, denoted as DR-KLD-MIG, will be comprehensively analyzed and compared against a suite of advanced algorithms. The comparison methods include the optimal processor under the complex Gaussian clutter assumption, STAP (geometry) [42]; the KLD-MIG detector without DR [28]; the displaced phase center antenna (DPCA) based on channel clutter data cancellation [48]; the robust PCA (RPCA) based on sparse and low-rank decomposition [49]; and the KLD-MIG detector using PCA for

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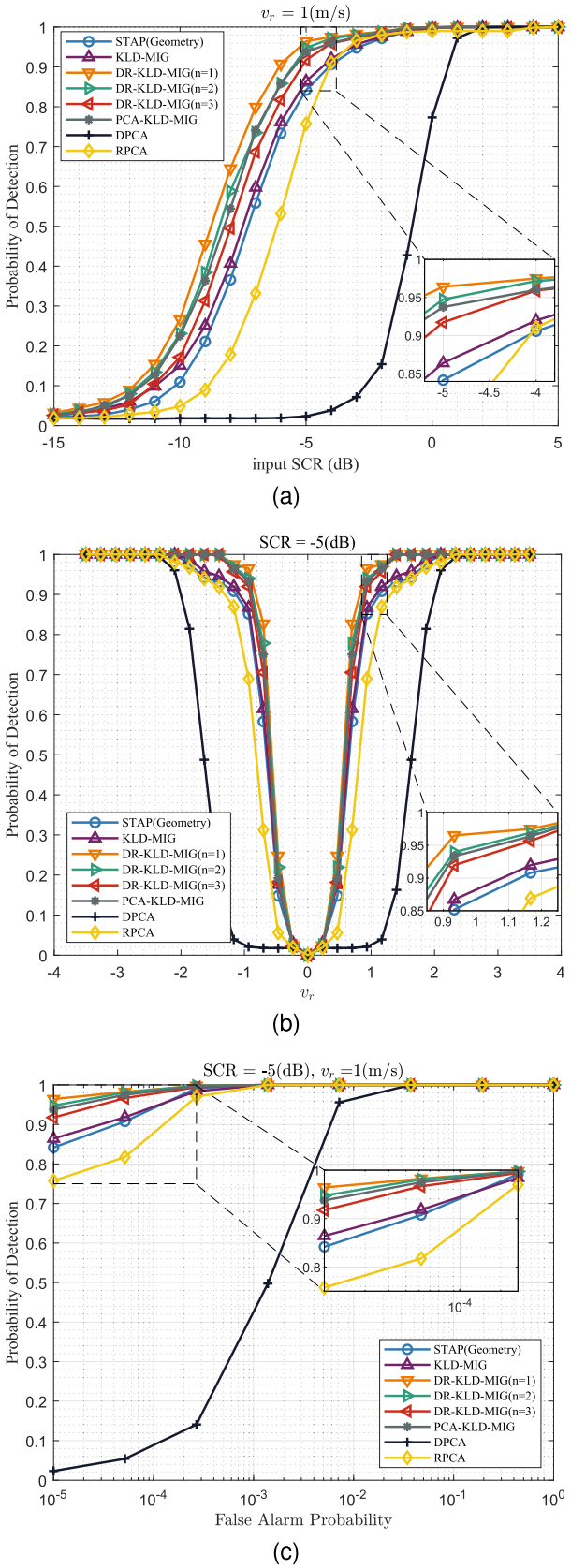


Fig. 8. Detection performance of the proposed method: (a) P_d versus SCR; (b) P_d versus v_r; and (c) ROC curve.

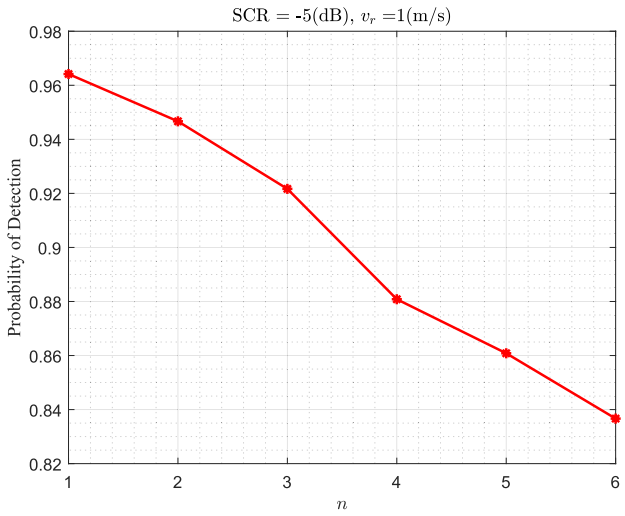


Fig. 9. Probabilities of detection versus dimension n.

linear feature DR [50]. Here, STAP (geometry) indicates that the CCM is estimated using the geometric mean of the clutter data. As has been found in [50], the PCA-based DR method only uses the CUT to construct the DR mapping and achieves the best performance when the dimensionality is $n = 1$. We adopted this setting in the comparison and denote it as PCA-KLD-MIG. To fully evaluate the detection performance of the proposed method, the false alarm probability is set as $P_{\text{FA}} = 10^{-5}$. Similarly, first fix the radial velocity at $v_r = 1$ m/s and vary the input SCR, then fix the input SCR at -5 dB and vary the radial velocity to evaluate the changes in the detection probability with the input SCR and the radial velocity, respectively. The evaluation range of the SCR is from -15 to 5 dB, and the evaluation range of the radial velocity is from -3.5 to 3.5 m/s. To more realistically simulate the processing flow of the real SAR system, the experiment of this study is based on the airborne multichannel SAR imaging geometry and signal model described in Section II-A. Starting from the scattering coefficients of the simulated scene, the entire working process of the SAR system, such as echo collection, imaging registration, clutter suppression, and moving target detection, is simulated. It is worth noting that since the multichannel interference phase is proportional to the radial velocity, generally, an increase in the radial velocity will lead to an increase in the detection probability. However, this is under the premise that the radial velocity should still be within the unambiguous radial velocity search space, that is, $|v_r| < \lambda V/2d$. If it exceeds this range, the interference phase along the trajectory will be ambiguous, and the detection probability will decrease. In this experiment, the unambiguous radial velocity range is $(-4.68, 4.68$ m/s), and the set radial velocity variation range $(-3.5, 3.5$ m/s) in the experiment is within this range. In the experiment, 12 moving targets are uniformly arranged in the scene, the number of training samples is 40, the training samples are taken from the adjacent area of the unit under test, and the protection unit is 5×11. After 100 independent Monte Carlo experiments, the obtained results are as given in Fig. 8.

From the experimental results, it can be seen that when the radial velocity is constant, as the input SCR increases,

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the detection probability of each method gradually increases. Under the same input SCR, the detection performances of STAP (geometry) and KLD-MIG are similar. The detection performance of both DPCA and RPCA is lower than that of STAP (geometry). The detection probabilities of the KLD-MIG methods after DR, including PCA-KLD-MIG and DR-KLD-MIG, are significantly improved compared with KLD-MIG. It can be observed from the experimental results that the smaller the dimension after DR of the DR-KLD-MIG method, the higher the detection probability. However, PCA-KLD-MIG is a linear DR method that fails to take into account the nonlinear structure of the signal itself. Moreover, it only uses the characteristics of CUT to construct the DR mapping without considering the characteristics of the clutter. Therefore, its detection performance is still slightly lower than that of the DR-KLD-MIG ($n = 1$) method. DR-KLD-MIG fully takes into account the nonlinear structure of the signal itself and aims to maximize KLD, thus achieving superior detection performance. When the input SCR remains unchanged, the detection probability of each method increases with the increase in radial velocity. Among them, DR-KLD-MIG ($n = 1$) has the narrowest notch, the smallest detectable velocity, and the optimal detection performance.

3) Analysis of Reduction Dimension: According to the previous discussion in Section IV, selecting DR enhancement mappings with different dimensions n will lead to significant differences in the detection performance of KLD-MIG. First, we further present the variation of detection probability with the DR dimension n when SCR $= -5$ dB and $v_r = 1$ m/s, as shown in Fig. 9.

It can be seen from Figs. 7 to 9 that the detection performance is optimal when the reduction dimension is 1, and the detection performance decreases as the reduction dimension increases. Regarding the selection of the optimal dimension, we provide the following theoretical explanation.

After prewhitening processing, the clutter covariance component in the original $\boldsymbol{R}_1$ is transformed into $\tilde{\boldsymbol{R}}_0 = \boldsymbol{I}$, whose eigenvalues are all 1. In contrast, the target covariance component after whitening processing becomes $\tilde{\boldsymbol{R}}_t = \boldsymbol{R}_0^{-1/2}\boldsymbol{R}_t\boldsymbol{R}_0^{-1/2}$, with eigenvalues significantly deviating from 1. To achieve optimal dimension reduction, we aim to find a transformation matrix that projects the high-dimensional HPD matrix into a low-dimensional submanifold, thereby maximizing the feature difference between the two hypotheses. From the perspective of detection theory, the optimal projection should cover all the target-related components and exclude dimensions that only contain whitened clutter (whose eigenvalues are 1 and provide no useful information for distinguishing targets). This leads to a key principle: the dimension n of the optimal projection must match the discrete bandwidth of the target spatial power spectrum. If we choose $n < B$, we would be discarding some of the target's energy, leading to a suboptimal test statistic; If we choose $n > B$, we would start to include dimensions containing only pure, whitened clutter. This would dilute the test statistic, effectively reducing the signal-to-clutter ratio in the feature space and thus degrading the detection performance.

The conclusion that the optimal dimension selection is equivalent to the discrete spatial power spectrum can also be explained from the perspective of the eigenvalue problem of the whitening covariance matrix.

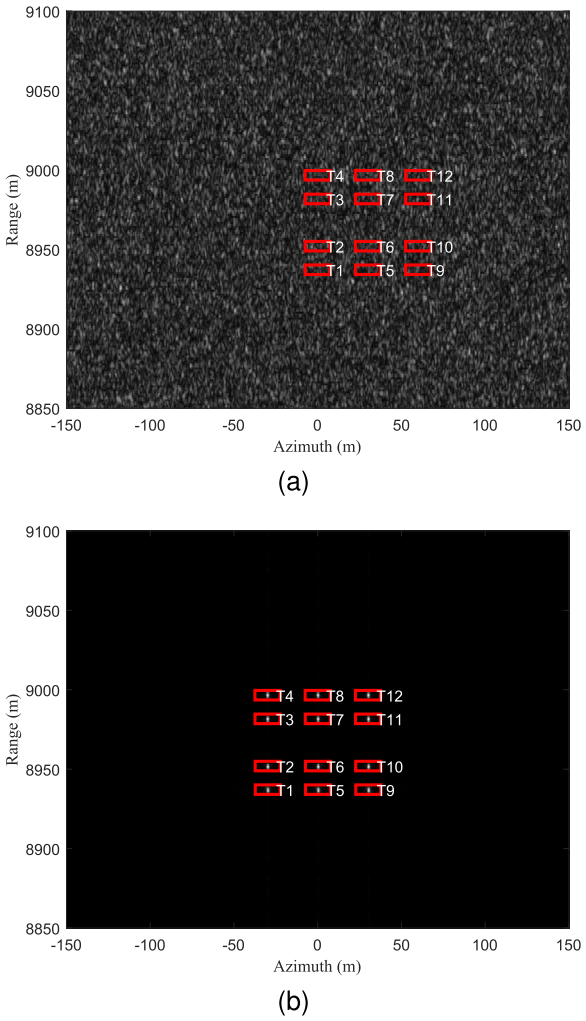


Fig. 10. Scene image in simulated clutter and target image: (a) scene image and (b) target image.

The KLD before DR can be expressed as

$$\mathcal{D}_{\mathrm{KL}}\left(\tilde{\boldsymbol{R}}_1, \boldsymbol{I}\right) = \sum_{i=1}^{N} \lambda_i - \log \lambda_i - 1. \tag{32}$$

According to the relationship between eigenvalues and power spectrum, the eigenvalue λ_k corresponds to the kth frequency component of the spatial power spectrum. Assume that the discrete spatial power spectrum of the target signal is B, that is, the effective width of its discrete Fourier transform. In the summation formula, among the N summation terms, only B terms contain the frequency components of the target signal, while the remaining $N - B$ terms are pure clutter. Therefore, when the dimension is exactly B, the detection statistic is the sum of all the spectral components of the target signal. This indicates that the dimension n should be consistent with the discrete bandwidth of the target signal.

Therefore, by setting $n = B$, the projection ensures that the resulting test statistic effectively becomes an integration (or sum) of all the target's spectral power components. This maximizes the discrimination between the two hypotheses.

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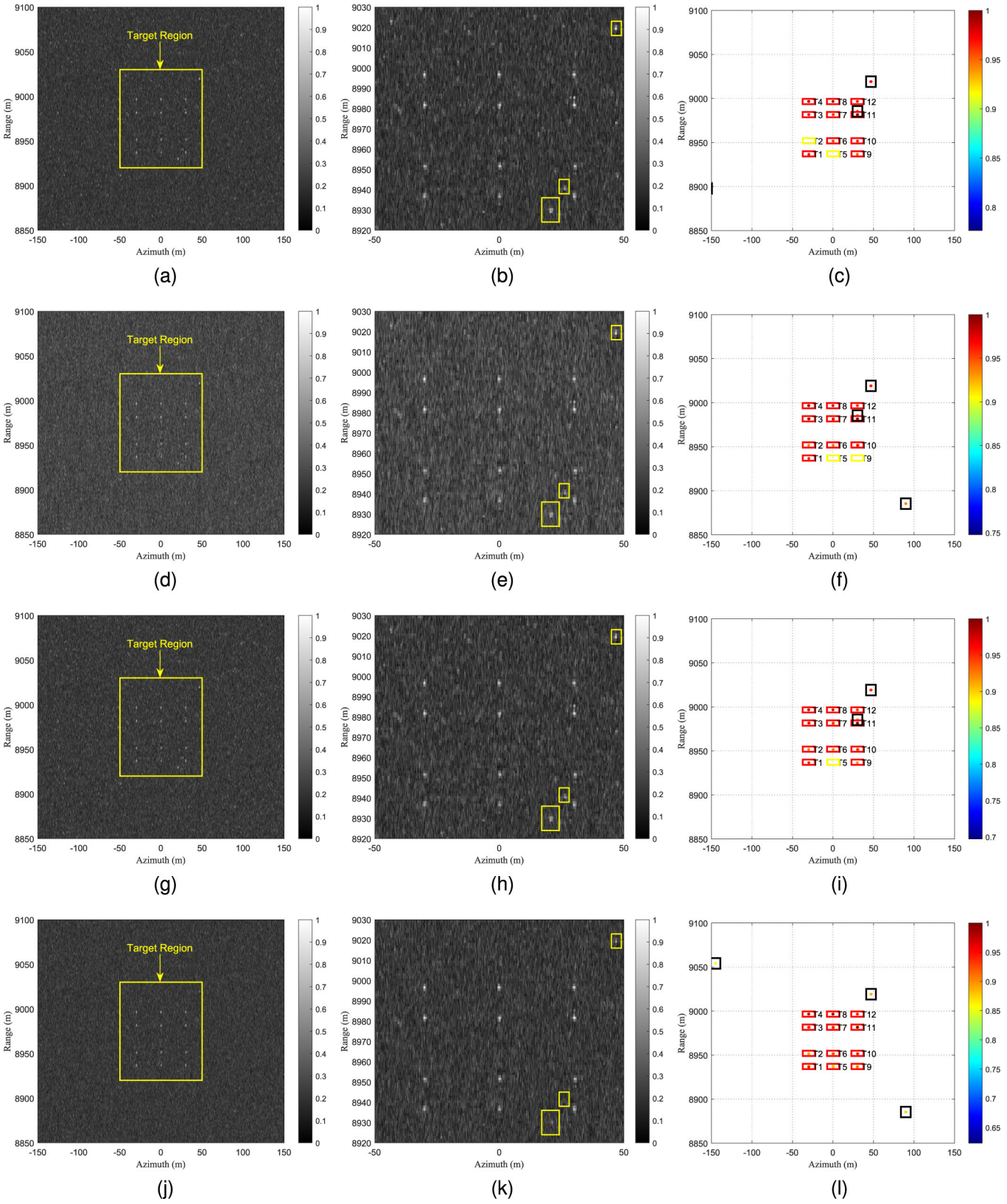


Fig. 11. Detection results under simulated clutter: (a) normalized test statistic of STAP (geometry); (b) normalized test statistic of target region by STAP (geometry); (c) detection result of STAP (geometry); (d) normalized test statistic of KLD-MIG; (e) normalized test statistic of target region by KLD-MIG; (f) detection result of KLD-MIG; (g) normalized test statistic of PCA-KLD-MIG; (h) normalized test statistic of target region by PCA-KLD-MIG; (i) detection result of PCA-KLD-MIG; (j) normalized test statistic of DR-KLD-MIG (n = 1); (k) normalized test statistic of target region by DR-KLD-MIG (n = 1); and (l) detection result of DR-KLD-MIG (n = 1).

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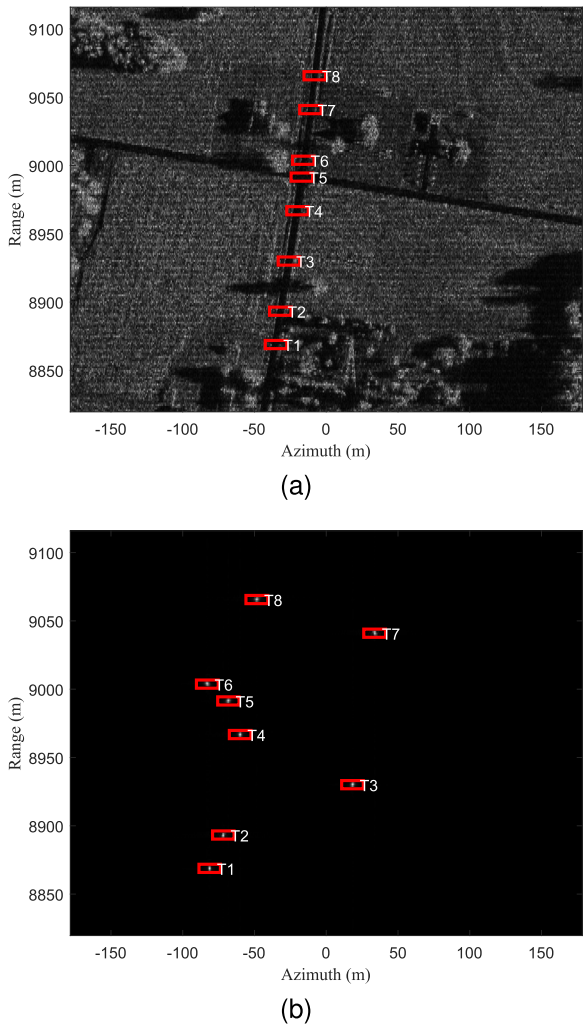


Fig. 12. Scene image in real clutter and target image: (a) scene image and (b) target image.

For the multichannel SAR constant velocity moving target model considered in our article, the target's spatial steering vector after channel registration in the image domain is given by (2), and (2) can be rewritten as

$$\begin{aligned} \boldsymbol{s} &= \left[1, \exp\left(j\frac{2\ \pi\ d_2\ v_r}{\lambda V}\right), \ldots, \exp\left(j\frac{2\ \pi\ d_N\ v_r}{\lambda V}\right)\right]^T \\ &= \left[1, \exp\left(j\ 2\pi\tilde{f}_s\right), \ldots, \exp\left(j\ 2\pi\,(N-1)\,\tilde{f}_s\right)\right]^T \end{aligned} \tag{33}$$

where $\tilde{f}_s = dv_r/\lambda V$. By performing an FFT along the spatial domain, we obtain a single spectral peak, which indicates that the spatial power spectrum of the target signal vector has a discrete bandwidth of $B = 1$. Therefore, projecting the high-dimensional HPD matrix into a 1-D submanifold that maximizes the feature differences maximizes the detection probability. This explains why $n = 1$ is the optimal choice for DR.

C. Detection Results in Simulated Clutter

In this section, the moving target detection results in the simulated clutter scene are presented. According to the discussion in Section IV-B, the proposed method DR-KLD-MIG has optimal performance when the DR dimension is $n = 1$. Meanwhile, based on the performance analysis of the detection in Section IV-B, among several comparison algorithms, STAP (geometry), KLD-MIG, and PCA-KLD-MIG are relatively superior. Therefore, in the experiments of this section and Section IV-D, the processing results of the proposed method DR-KLD-MIG ($n = 1$) are compared with the classical STAP (geometry), KLD-MIG, and the PCA-KLD-MIG. The system simulation parameters still use the parameters shown in Table I. Twelve moving targets are uniformly arranged in the scene, and the target motion parameters are all set as $v_x = 0$ m/s, $v_r = 1$ m/s, and the scene SCR is -5 dB. Fig. 10(a) and (b) provides the SAR image in the clutter background and the SAR image of moving targets in the absence of clutter, respectively. The red box marks the initial layout position of the moving targets in the scene and the real imaging position of the targets. It can be seen that due to the influence of the radial velocity, the Doppler center of the moving target deviates from the zero frequency, and the imaging result has a certain deviation in the azimuth direction compared with the layout position.

TABLE II
PARAMETERS OF MOVING TARGETS

Target	$x(m)$	$R_B(m)$	$v_x(m/s)$	$v_r(m/s)$
T1	-35	8869	2.05	1.54
T2	-32	8893	2.33	1.33
T3	-26	8930	2.81	-1.48
T4	-20	8967	1.70	1.34
T5	-17	8991	-3.21	1.71
T6	-16	9004	2.43	2.23
T7	-11	9041	1.00	-1.49
T8	-8	9066	1.51	1.35

Fig. 11(a), (d), (g), and (j), respectively, presents the normalized test statistics of each method. The area where the target is located is marked by a yellow rectangular box. The enlarged results of the target region after the processing of each method are shown in Fig. 11(b), (e), (h), and (k), respectively. In the enlarged results of the target area, several typical strong discrete clutter residues are marked by the yellow box. The STAP (geometry) method is the optimal filter under the complex Gaussian assumption. However, the heterogeneous clutter environment at this time not only significantly deviates from the complex Gaussian distribution but also there are discrete clutters with strong energy in the scene. At this time, STAP (geometry) is difficult to suppress such discrete clutter. Although the KLD-MIG method does not rely on a specific statistical distribution, due to the lack of DR processing, the geometric distance between the strong discrete clutter and the target on the HPD manifold is small, so its ability to suppress such strong discrete clutters is limited. Compared with KLD-MIG, the PCA-KLD-MIG method shows improved processing performance. However, its DR mapping only relies on the construction of the CUT and does not use the characteristics of clutter. Its ability to suppress strong discrete clutter is still limited. In the processing result of DR-KLD-MIG ($n = 1$), since the DR mapping processing further increases the geometric

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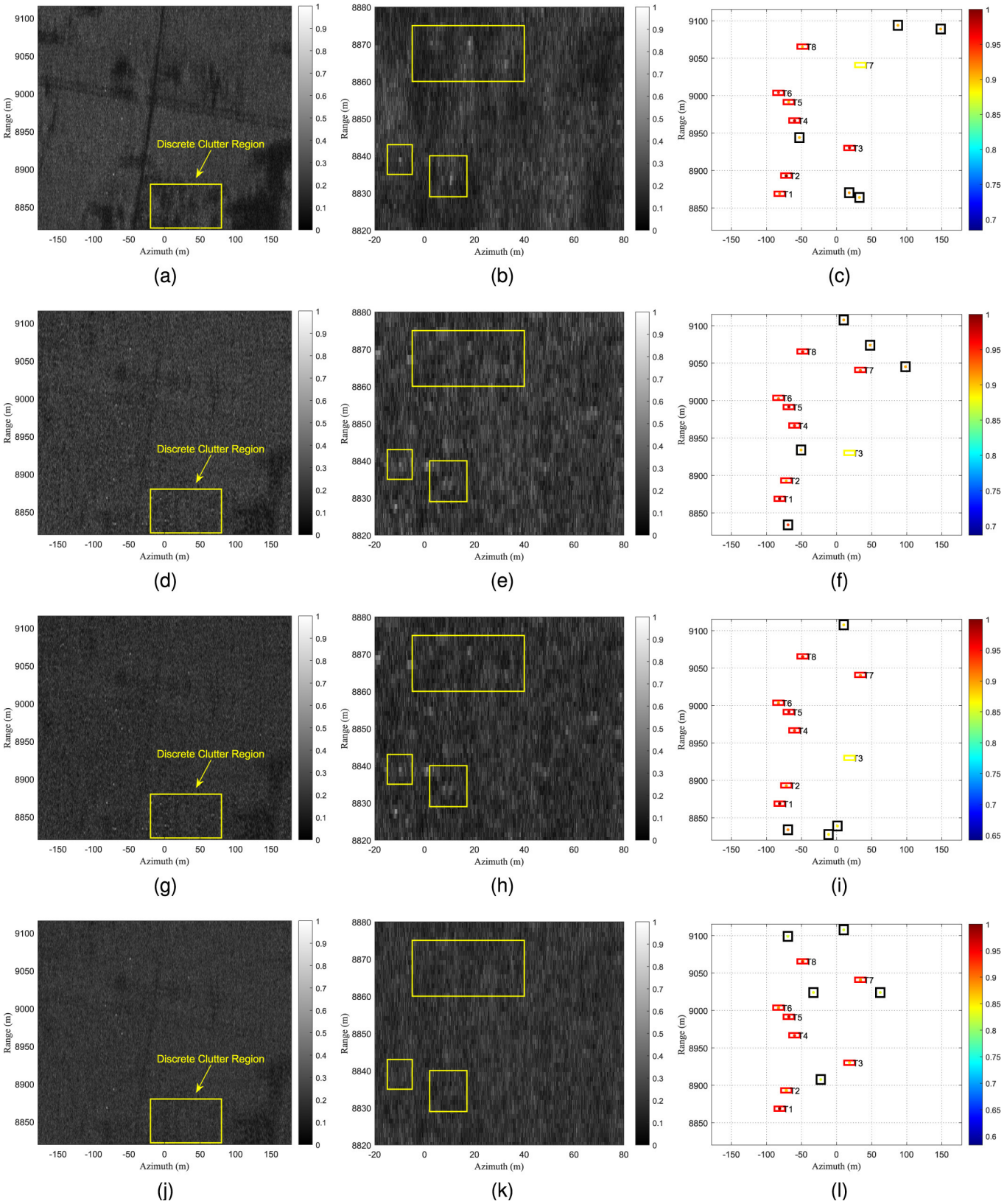


Fig. 13. Detection results under real clutter: (a) normalized test statistic of STAP (geometry); (b) normalized test statistic of discrete clutter region by STAP (geometry); (c) detection result of STAP (geometry); (d) normalized test statistic of KLD-MIG; (e) normalized test statistic of discrete clutter region by KLD-MIG; (f) detection result of KLD-MIG; (g) normalized test statistic of PCA-KLD-MIG; (h) normalized test statistic of discrete clutter region by PCA-KLD-MIG; (i) detection result of PCA-KLD-MIG; (j) normalized test statistic of DR-KLD-MIG ($n = 1$); (k) normalized test statistic of discrete clutter region by DR-KLD-MIG ($n = 1$); and (l) detection result of DR-KLD-MIG ($n = 1$).

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distance between the strong discrete clutter and the moving target, these strong discrete clutters are well-suppressed, and it has superior processing performance. It is worth noting that KLD-MIG lacks a target steering vector search step and does not perform precise matching for target signals, resulting in relatively high residual clutter power. Fig. 11(c), (f), (i), and (l) presents the detection results of each method, where the detection threshold is determined by the false alarm probability $P_{\mathrm{FA}} = 10^{-5}$. Among them, the red box represents correct detection, the yellow box represents missed detection, and the black box represents false alarm. From the detection results, it can be seen that there are two missed detections in the STAP (geometry) and KLD-MIG methods, one missed detection in the PCA-KLD-MIG method, while DR-KLD-MIG ($n = 1$) correctly detects all the moving targets and has the best detection performance.

D. Detection Results in Measured Clutter

This section provides the detection results based on the measured clutter background. The depicted clutter scene is a representative heterogeneous environment featuring trees and roads, with eight moving targets arrayed along the road. The SCR of this scene is −6 dB. The parameters of the moving targets set in this experiment are shown in Table II. Fig. 12(a) and (b) provides the SAR image in the clutter background and the SAR image of moving targets in the absence of clutter, respectively.

By analyzing the normalized test statistics of each method, it can be seen that the KLD-MIG method lacks a target steering vector search step and does not perform precise matching for target signals, resulting in high residual clutter power. Although the STAP (geometry) method has less residual clutter compared with the KLD-MIG method, it cannot suppress the strong discrete clutter in the scene. Compared with KLD-MIG, the PCA-KLD-MIG method shows improved processing performance, but its ability to suppress strong discrete clutter is limited. The DR-KLD-MIG ($n = 1$) processing result shows the least clutter residue and the strongest target energy, while effectively suppressing strong discrete clutter in the scene. In the marked area with a yellow box in the displayed scene, the trees are relatively dense, generating a lot of strong discrete clutter that is difficult to handle by traditional methods. To more clearly see the advantages of the proposed method, the enlarged results of this area in the processing of each method are given. There are several discrete clutters with large scattering intensities within the yellow box. The STAP (geometry) method, KLD-MIG method, and PCA-KLD-MIG method are difficult to effectively suppress such clutters. In the processing results of DR-KLD-MIG ($n = 1$), such strong discrete clutters are well-suppressed, and the residual clutter energy is the lowest, demonstrating the superior processing performance. It is worth noting that in the processing results of the KLD-MIG, PCA-KLD-MIG, and DR-KLD-MIG ($n = 1$) methods, there are fewer residual edges of trees and road contours; in contrast, the processing results of the STAP (geometry) method exhibit obvious residual edges of such complex terrain clutter. This indicates that the IG detector demonstrates better robustness in heterogeneous complex clutter scenarios. Fig. 13(c), (f), (i), and (l) presents the detection results of each method, and the false alarm probability is still set as $P_{\mathrm{FA}} = 10^{-5}$. From the detection results, it can be seen that there is one missed detection in the STAP (geometry), KLD-MIG, and PCA-KLD-MIG, and DR-KLD-MIG ($n = 1$) correctly detects all the moving targets and has the best detection performance.

V. Conclusion

This article focused on the problem of clutter suppression and moving target detection of airborne multichannel SAR in a heterogeneous clutter environment and proposed an HPD manifold DR mapping method based on the KLD to achieve the enhanced detection of multichannel SAR moving targets. First, the airborne multichannel SAR image domain data are characterized as points on the HPD manifold through feature mapping to construct a feature manifold. Subsequently, we construct an optimal DR mapping optimization problem based on maximizing the KLD on the HPD manifold and derive the analytical solution of this optimization problem, avoiding the huge computational cost brought using the Riemannian conjugate gradient method. Based on this analytical solution, we construct the optimal DR matrix and project the data into the low-dimensional HPD manifold. Finally, on the dimensionality-reduced HPD manifold, the moving target enhanced detection is realized using KLD-MIG. The proposed multichannel SAR moving target enhanced detection framework based on KLD DR mapping effectively avoided the computational cost caused using the Riemannian conjugate gradient method and significantly reduced the computational complexity, making it suitable for the processing of the SAR system. The experimental results show that the distinguishability between the target and the clutter on the original HPD manifold is significantly reduced in the scenarios of low SCR and slow-moving targets, while the proposed DR mapping method can effectively increase the distinguishability between the target and the clutter on the HPD manifold, especially in the scenarios of low SCR and slow-moving targets. Compared with the traditional STAP method (using the geometric mean to estimate the covariance matrix) and KLD-MIG, the proposed method shows more superior clutter suppression ability (especially for strong discrete clutter) and better detection performance in the heterogeneous clutter environment background.

Appendix
Proof of Proposition 1

Proof: Since $\tilde{\boldsymbol{R}}_1$ is an HPD matrix, for $\forall \boldsymbol{u} \in \mathbb{C}^N, \boldsymbol{u} \neq \boldsymbol{0}$, we have $\boldsymbol{u}^H \tilde{\boldsymbol{R}}_1 u > 0$. Then, for $\forall \boldsymbol{v} \in \mathbb{C}^N, \boldsymbol{v} \neq \boldsymbol{0}$, $\boldsymbol{v}^H \boldsymbol{Q}^H \boldsymbol{R}_1 \tilde{Q} v = (\boldsymbol{Q}\boldsymbol{v})^H \tilde{\boldsymbol{R}}_1 (\boldsymbol{Q}\boldsymbol{v})$. Since $\boldsymbol{Q}^H \boldsymbol{Q} = \boldsymbol{I}_n$ and $\boldsymbol{Q}$ is a column-full-rank matrix, when $\boldsymbol{v} \neq \boldsymbol{0}$, $\boldsymbol{Q}\boldsymbol{v} \neq \boldsymbol{0}$. Thus, $\boldsymbol{Q}^H \boldsymbol{R}_1 \tilde{Q}$ is also an HPD matrix. Let its eigenvalues be arranged in descending order as $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n > 0$.

To prove the constraint relationship between the eigenvalues of $\tilde{\boldsymbol{R}}_1$ and $\boldsymbol{Q}^H \boldsymbol{R}_1 \tilde{Q}$, we first use the Courant–Fischer theorem [51] to give the variational representation of the eigenvalues of the Hermitian matrix.

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For the kth largest eigenvalue λ_k of $\tilde{\boldsymbol{R}}_1$

$$\lambda_k = \max_{\substack{\mathcal{S}\subset\mathbb{C}^N \\ \dim(\mathcal{S})=k}} \min_{\substack{\boldsymbol{x}\in\mathcal{S} \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x} \tag{34}$$

$$\lambda_k = \min_{\substack{\mathcal{T}\subset\mathbb{C}^N \\ \dim(\mathcal{T})=N-k+1}} \max_{\substack{\boldsymbol{x}\in\mathcal{T} \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x}. \tag{35}$$

For the kth largest eigenvalue μ_k of $\boldsymbol{Q}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{Q}$

$$\mu_k = \max_{\substack{\mathcal{V}\subset\mathbb{C}^n \\ \dim(\mathcal{V})=k}} \min_{\substack{\boldsymbol{y}\in\mathcal{V} \\ \|\boldsymbol{y}\|=1}} \boldsymbol{y}^H \boldsymbol{Q}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{Q} \boldsymbol{y} \tag{36}$$

$$\mu_k = \min_{\substack{\mathcal{W}\subset\mathbb{C}^n \\ \dim(\mathcal{W})=n-k+1}} \max_{\substack{\boldsymbol{y}\in\mathcal{W} \\ \|\boldsymbol{y}\|=1}} \boldsymbol{y}^H \boldsymbol{Q}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{Q} \boldsymbol{y}. \tag{37}$$

First, we prove $\mu_k \le \lambda_k$: consider the max–min expression of μ_k. Let $\mathcal{V}_k$ be the k-dimensional subspace in $\mathbb{C}^n$ that achieves the maximum value. Let $\mathcal{S}_k = \boldsymbol{Q}\mathcal{V}_k = \{\boldsymbol{Q}\boldsymbol{y} \mid \boldsymbol{y} \in \mathcal{V}_k\}$. Since the columns of $\boldsymbol{Q}$ are linearly independent (in fact, orthonormal), $\mathcal{S}_k$ is a k-dimensional subspace in $\mathbb{C}^N$. Meanwhile, for $\boldsymbol{y} \in \mathbb{C}^n$, let $\boldsymbol{x} = \boldsymbol{Q}\boldsymbol{y}$, then $\|\boldsymbol{x}\|^2 = (\boldsymbol{Q}\boldsymbol{y})^H(\boldsymbol{Q}\boldsymbol{y}) = \boldsymbol{y}^H\boldsymbol{Q}^H\boldsymbol{Q}\boldsymbol{y} = \boldsymbol{y}^H\boldsymbol{I}_n\boldsymbol{y} = \|\boldsymbol{y}\|^2$. Therefore, $\|\boldsymbol{y}\| = 1$ if and only if $\|\boldsymbol{x}\| = 1$.

Then,

$$\mu_k = \min_{\substack{\boldsymbol{y}\in\mathcal{V}_k \\ \|\boldsymbol{y}\|=1}} \boldsymbol{y}^H \boldsymbol{Q}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{Q} \boldsymbol{y} = \min_{\substack{\boldsymbol{x}\in\mathcal{S}_k \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x}. \tag{38}$$

Now consider the max–min expression of λ_k

$$\lambda_k = \max_{\substack{\mathcal{S}\subset\mathbb{C}^N \\ \dim(\mathcal{S})=k}} \min_{\substack{\boldsymbol{x}\in\mathcal{S} \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x}. \tag{39}$$

Since $\mathcal{S}_k$ is just a specific k-dimensional subspace in $\mathbb{C}^N$, the minimum value on this subspace must be less than or equal to the maximum value of the minimum values on all k-dimensional subspaces.

Therefore,

$$\mu_k = \min_{\substack{\boldsymbol{x}\in\mathcal{S}_k \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x} \le \max_{\substack{\mathcal{S}\subset\mathbb{C}^N \\ \dim(\mathcal{S})=k}} \min_{\substack{\boldsymbol{x}\in\mathcal{S} \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x} = \lambda_k. \tag{40}$$

Thus, we prove $\mu_k \le \lambda_k$.

Next, we prove $\lambda_{N-n+k} \le \mu_k$: consider the min–max expression of μ_k. Let $\mathcal{W}_k$ be the $(n-k+1)$-dimensional subspace in $\mathbb{C}^n$ that achieves the minimum value. Let $\mathcal{T}_k = \boldsymbol{Q}\mathcal{W}_k = \{\boldsymbol{Q}\boldsymbol{y} \mid \boldsymbol{y} \in \mathcal{W}_k\}$. $\mathcal{T}_k$ is an $(n-k+1)$-dimensional subspace in $\mathbb{C}^N$.

Then,

$$\mu_k = \max_{\substack{\boldsymbol{y}\in\mathcal{W}_k \\ \|\boldsymbol{y}\|=1}} \boldsymbol{y}^H \boldsymbol{Q}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{Q} \boldsymbol{y} = \max_{\substack{\boldsymbol{x}\in\mathcal{T}_k \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x}. \tag{41}$$

Now consider the min–max expression of λ_{N-n+k}

$$\lambda_{N-n+k} = \min_{\substack{\mathcal{T}\subset\mathbb{C}^N \\ \dim(\mathcal{T})=n-k+1}} \max_{\substack{\boldsymbol{x}\in\mathcal{T} \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x}. \tag{42}$$

Since $\mathcal{T}_k$ is just a specific $(n-k+1)$-dimensional subspace in $\mathbb{C}^N$, the maximum value on this subspace must be greater than or equal to the minimum value of the maximum values on all $(n-k+1)$-dimensional subspaces.

Therefore,

$$\mu_k = \max_{\substack{\boldsymbol{x}\in\mathcal{T}_k \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x} \ge \min_{\substack{\mathcal{T}\subset\mathbb{C}^N \\ \dim(\mathcal{T})=n-k+1}} \max_{\substack{\boldsymbol{x}\in\mathcal{T} \\ \|\boldsymbol{x}\|=1}} \boldsymbol{x}^H \tilde{\boldsymbol{R}}_1 \boldsymbol{x} = \lambda_{N-n+k}. \tag{43}$$

This proves $\lambda_{N-n+k} \le \mu_k$.□

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